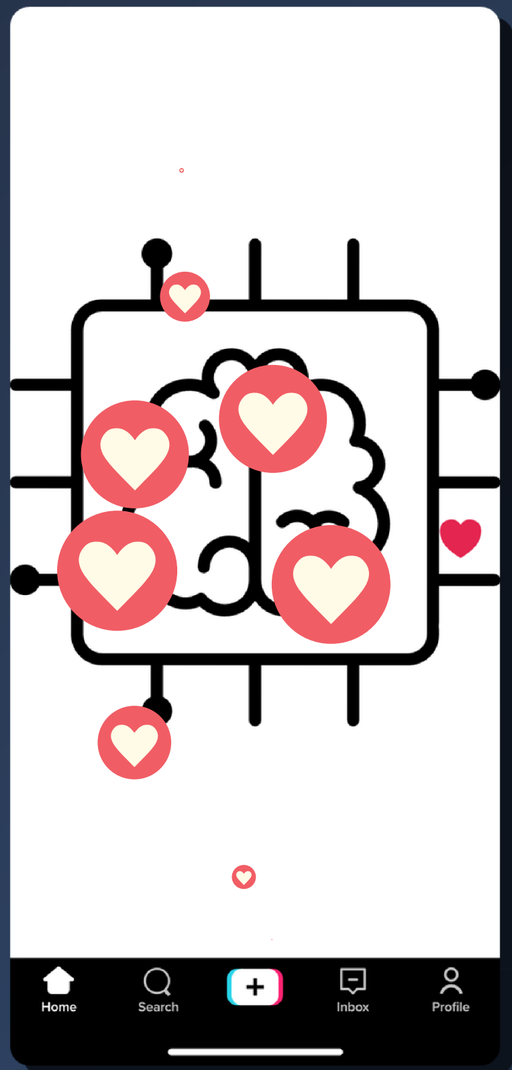
TikTok Creator Intelligence System

NLP-Based Comment Analysis & Content Insights

This project analyzes TikTok comments using Natural Language Processing to help small creators understand audience sentiment, identify key topics, and make data-driven content decisions.

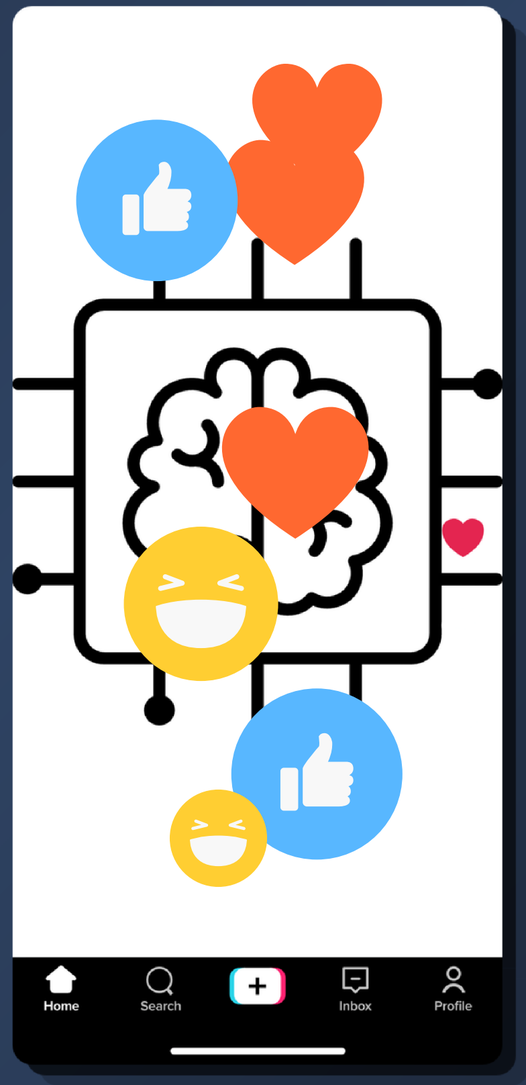
Turning Audience Feedback into Actionable Insights



Presented by Chinelo Lydia Nweke — IBS6 2026

10/04/2026

Deliverable 1
Problem & Goals



Problem Statement

Small TikTok creators struggle to extract meaningful insights from user comments because the feedback is unstructured, scattered, and difficult to analyze manually.

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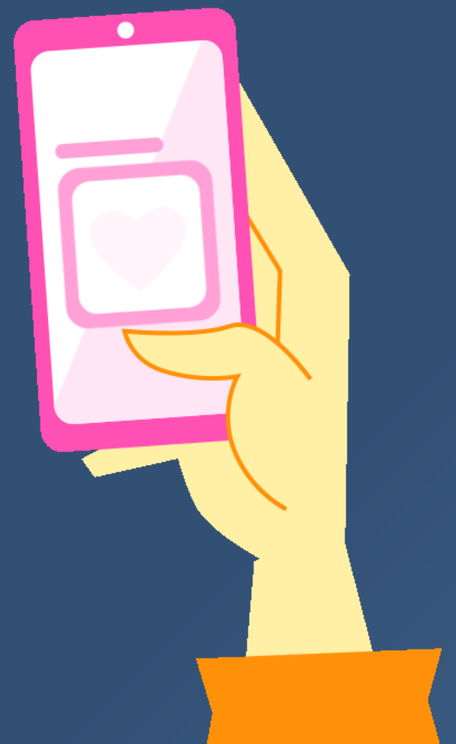
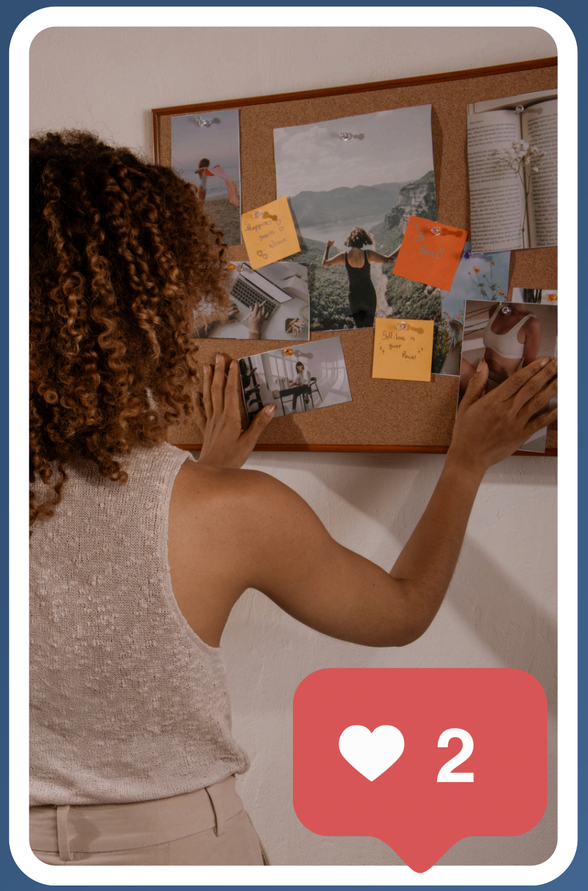
User Description

A small TikTok creator (1k–10k followers) who posts content regularly and wants to understand audience reactions but currently relies on manually reading comments and guessing what works.



They also struggle to decide which niche or content direction to focus on because they lack clear insight into what their audience consistently enjoys or engages with.



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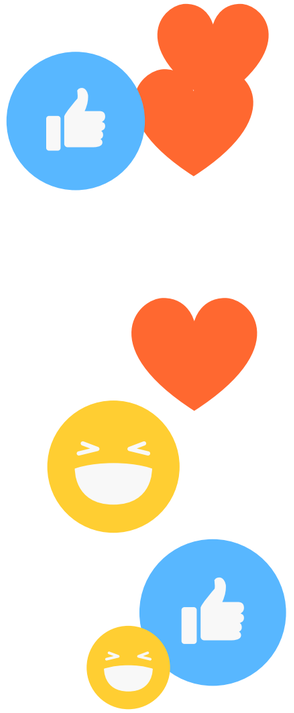
Why This Problem is Relevant

Comments contain valuable feedback such as opinions, questions, and suggestions, but manually analyzing them is time-consuming and inconsistent. As a result, creators miss patterns in audience preferences and lose opportunities to improve content and engagement.



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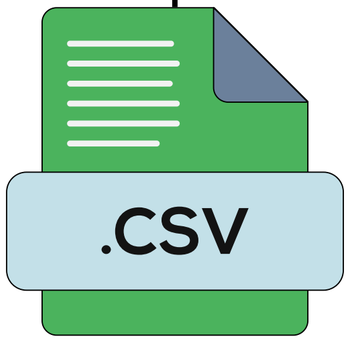
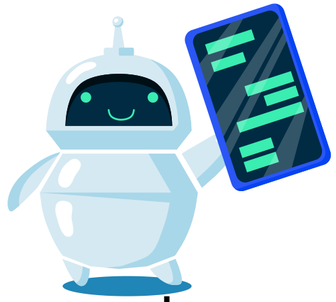
→ Sketch (Input → NLP Task → Output)

INPUT

CSV file containing TikTok comments and basic video metrics (likes, views, video type)



NLP Task

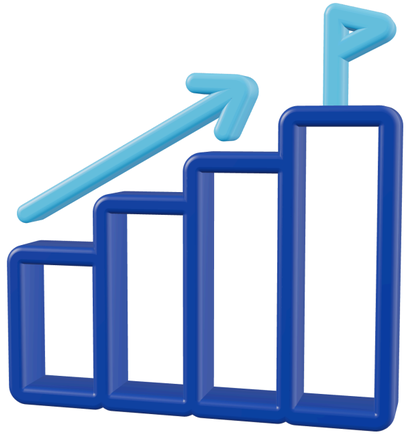


Text preprocessing
Sentiment analysis (positive, negative, neutral)
Keyword extraction (frequent topics and terms)

OUTPUT

- Sentiment distribution (e.g., 70% positive)
- Key topics discussed in comments
- Simple insights (what users like or complain about)
- Basic recommendations for content improvement

SMART Goals



Main Goal
By July 2026, develop a web-based NLP prototype that analyzes TikTok comments from a CSV file and generates sentiment summaries, keyword insights, and basic content recommendations within a few seconds for datasets of up to 300 comments.

Sub-Goals
Implement a preprocessing pipeline to clean and normalize comment text
Build a sentiment analysis module that classifies comments into positive, negative, and neutral
Develop a simple web interface (Streamlit) for uploading data and displaying insights

Non-Goals
No integration with TikTok API (data will be provided via CSV)
No training of custom deep learning models
No mobile application development

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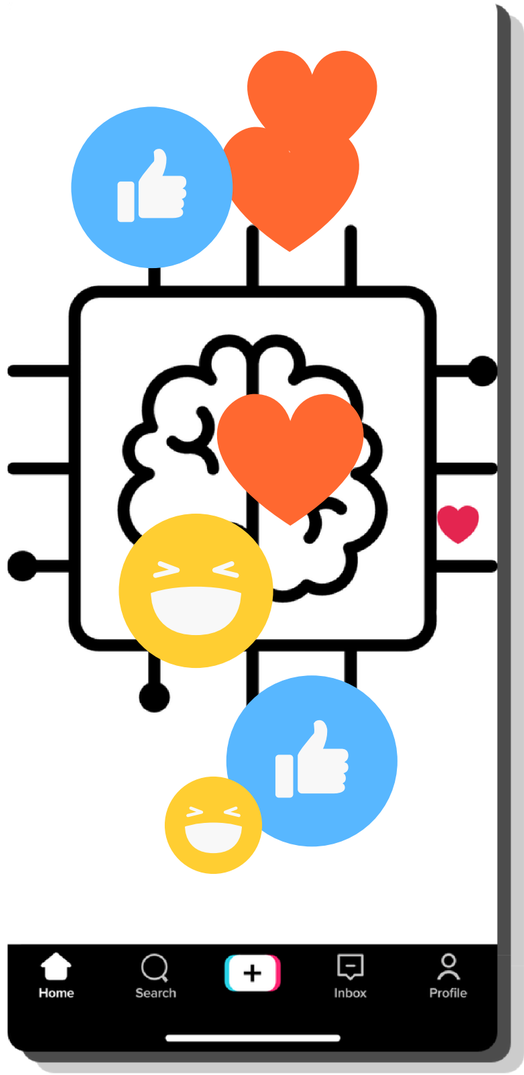
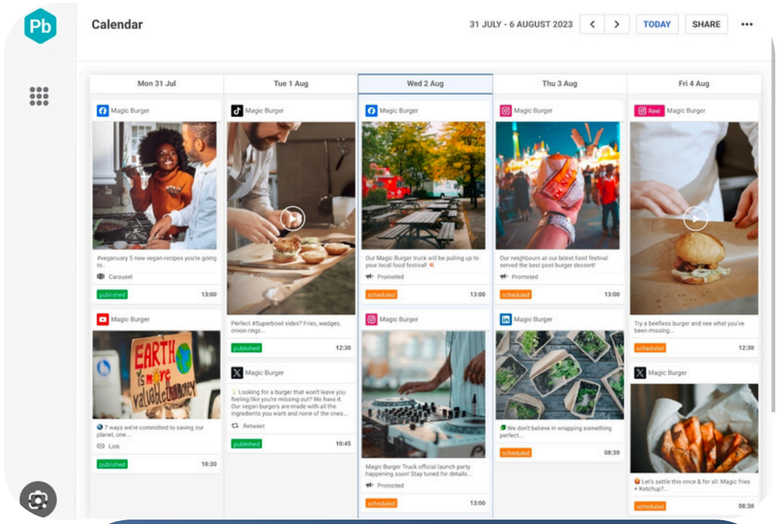
Deliverable 2
State of the Art

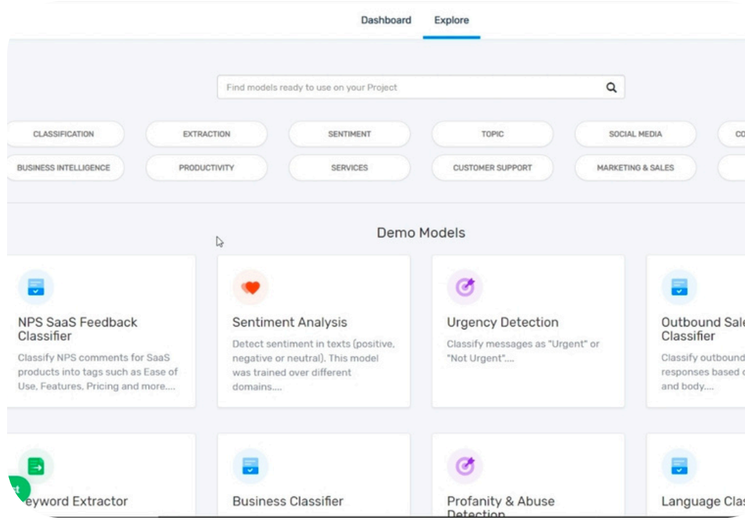


Existing Products / Prototypes

Hootsuite Insights
Social media monitoring and sentiment analysis
Tracks audience reactions and trends



Brandwatch
AI-based sentiment and trend analysis
Large-scale social media analytics



MonkeyLearn
Sentiment analysis and keyword extraction
General-purpose NLP tool

Limitations of Existing Solutions

- Designed for large companies, not small creators
- Expensive and not easily accessible
- Not tailored to TikTok-specific content
- Focus on data visualization, not actionable insights
- Do not connect comments directly to content strategy

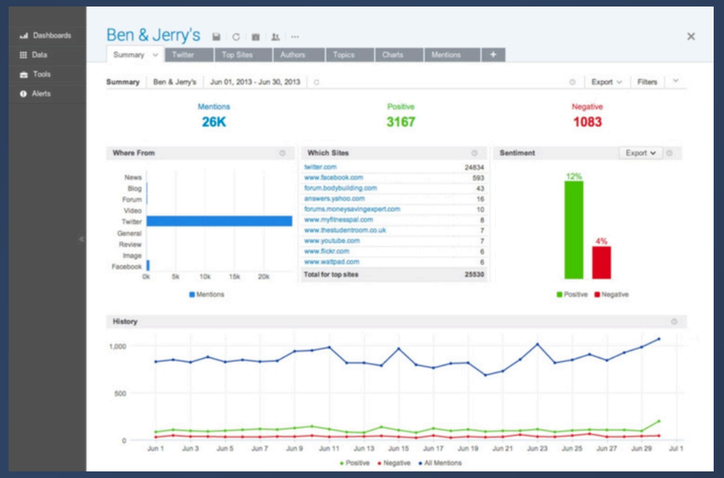
17/04/2026

17/04/2026

Reverse Engineering

Methods Used:
Sentiment Analysis (text classification: positive, negative, neutral)
Keyword Extraction (TF-IDF, frequency-based methods)

Models / Tools:
BERT (for text understanding)
spaCy / scikit-learn (NLP processing)



Interface:
Web dashboards for visualization and interaction

Parts I Can Reuse

- Text preprocessing pipeline (cleaning and normalization)
- Sentiment classification methods
- Keyword extraction techniques
- Basic dashboard structure for displaying results

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My Contribution (Delta)

- Focus on small TikTok creators (1k–10k followers)
- Simple and accessible system using CSV input
- Links comments to individual video performance
- Generates actionable insights, not just data
- Helps creators decide what content and niche to focus on

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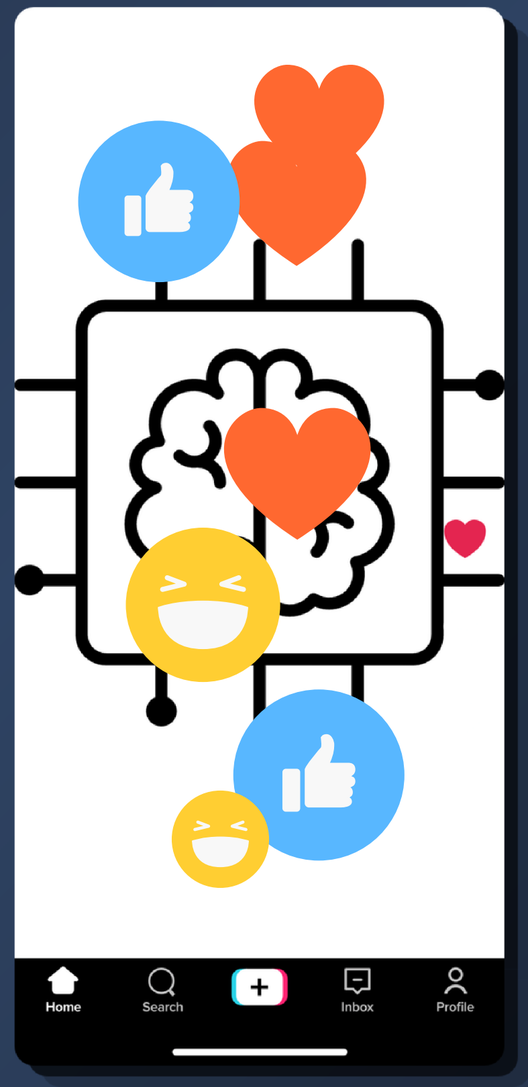
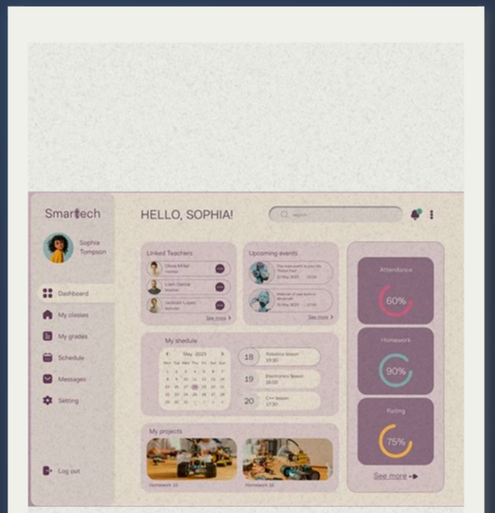
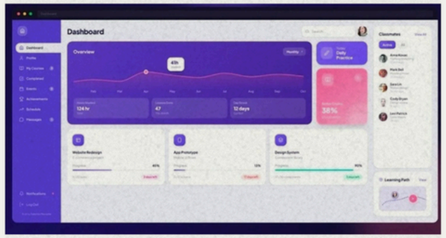
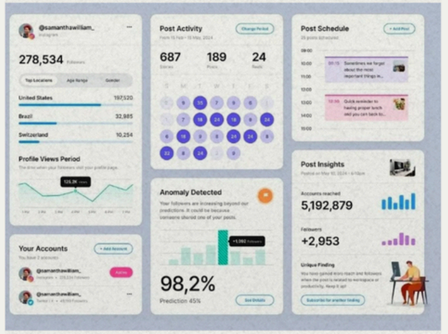
Deliverable 3
User Experience

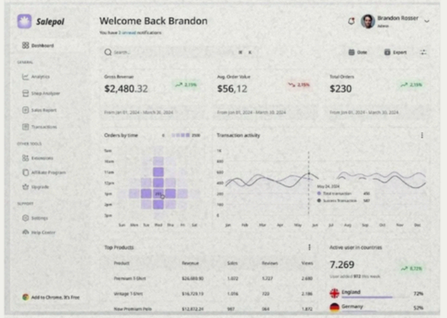


Moodboard

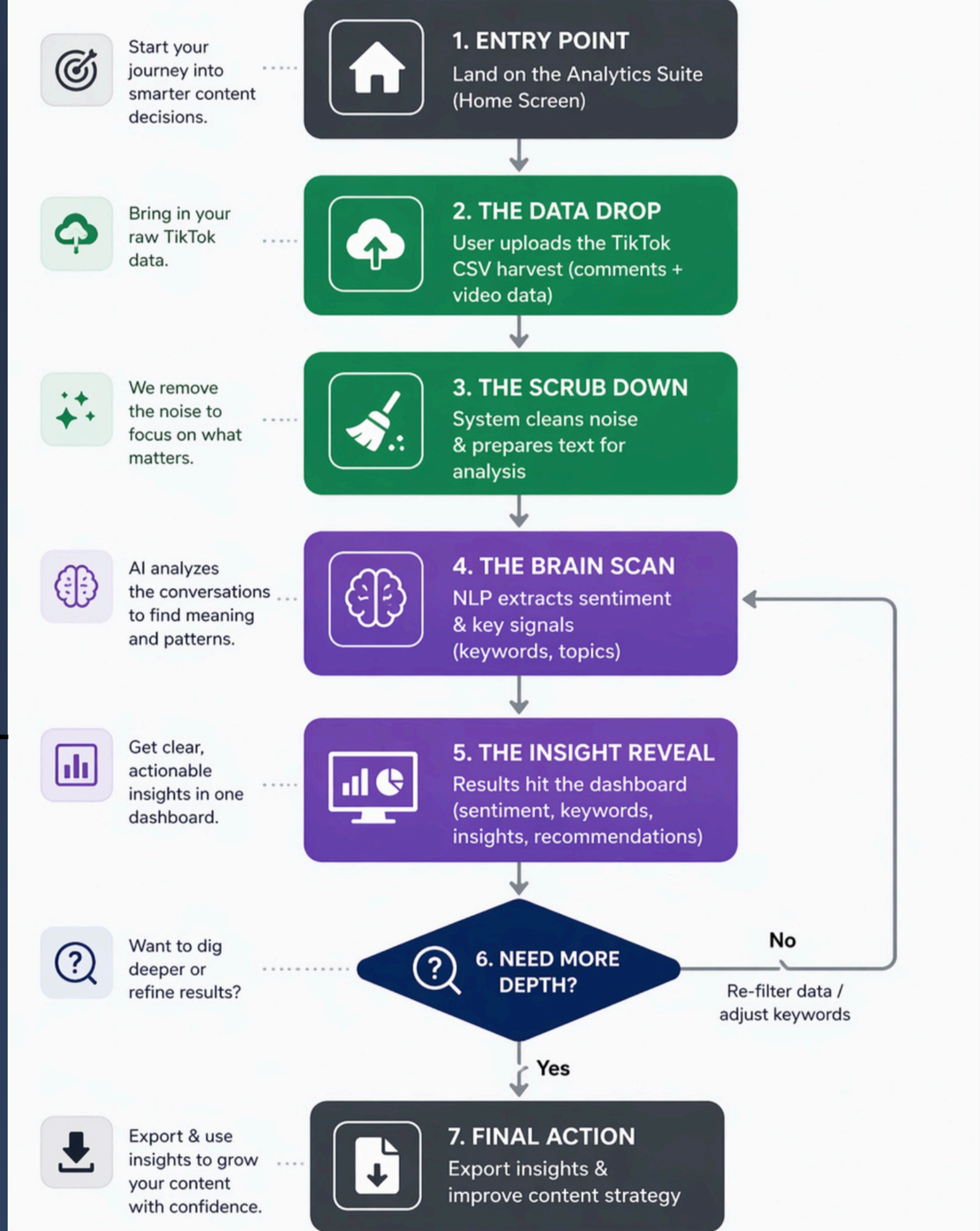

Smartech
HELLO, SOPHIA!
Sophia Tompson
Dashboard
My classes
My grades
Schedule
Messages
Setting
Log out
Linked Teachers
Upcoming events
Attendance
60%
Homework
90%
Rating
75%
My shedule
My projects
See more


Dashboard
Overview
Classmates
Daily Practice
Learning Path

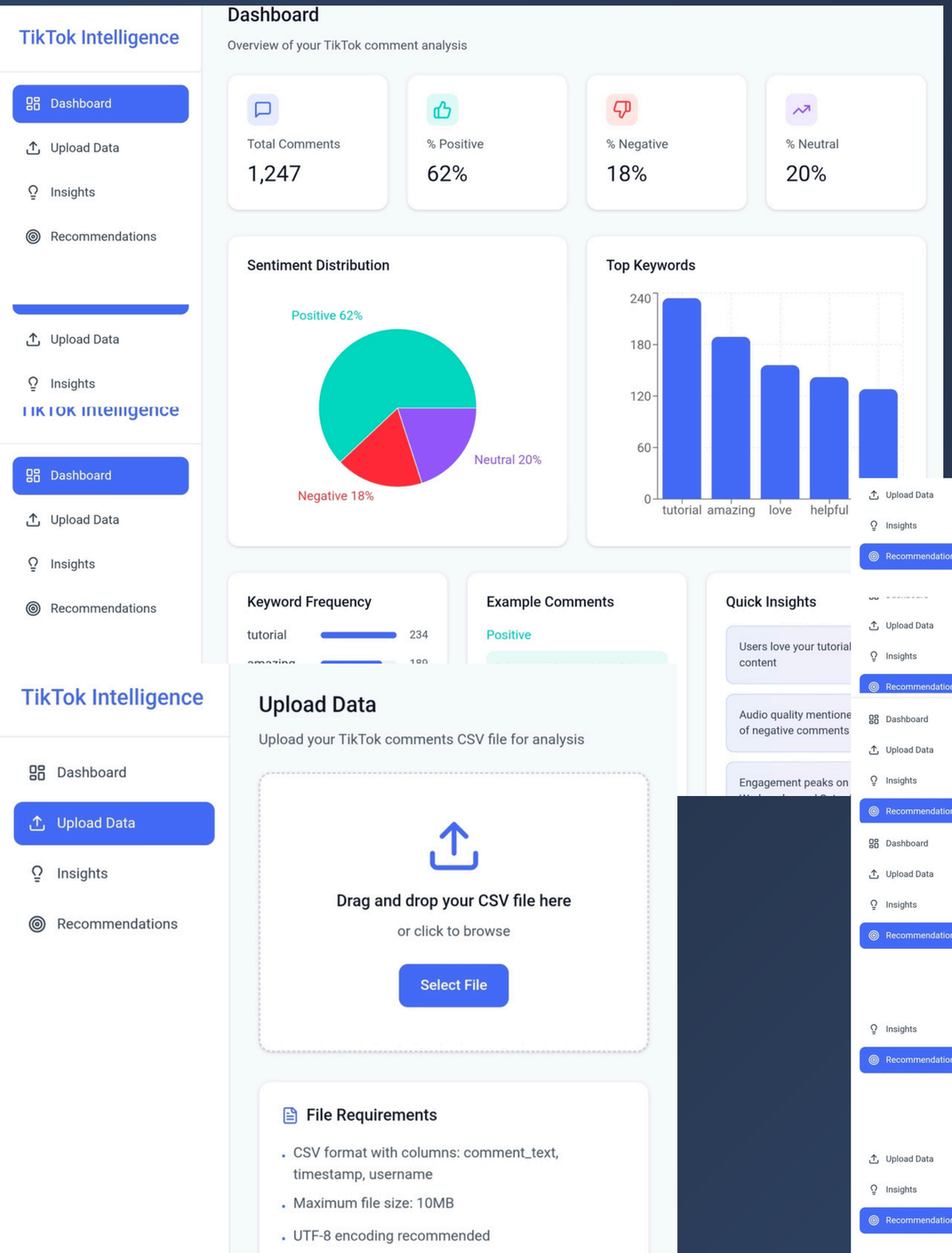

278,534
Profile Views Period
Post Activity
687
189
24
Post Schedule
Anomaly Detected
98,2%
Prediction 45%
Your Accounts
Post Insights
Accounts reached
5,192,879
Followers
+2,953
Unique Finding


Salepal
Welcome Back Brandon
Gross Revenue
$2,480.32
Avg. Order Value
$56.12
Total Orders
$230
Orders by time
Transaction activity
Top Products
Active user in countries
7.269

User Flow

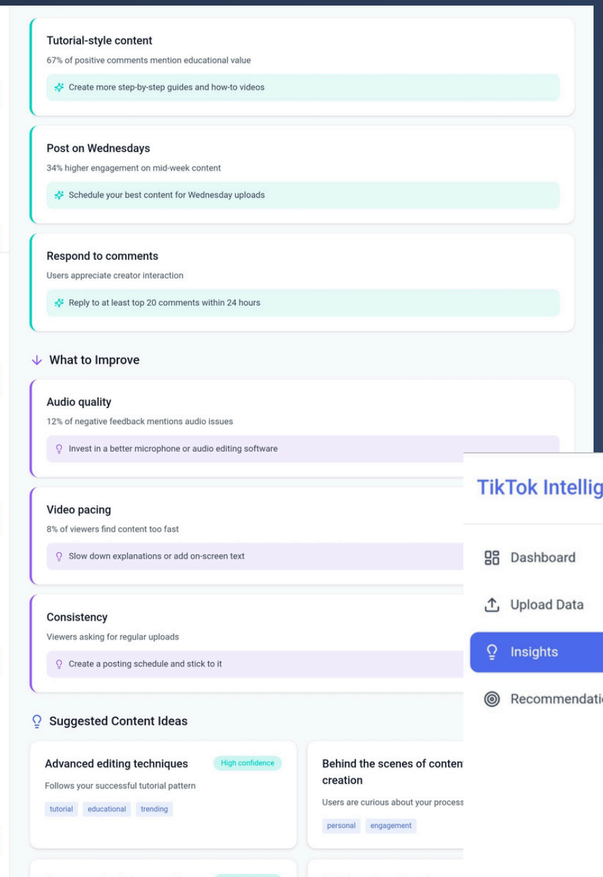

Start your journey into smarter content decisions.
1. ENTRY POINT
Land on the Analytics Suite (Home Screen)
Bring in your raw TikTok data.
2. THE DATA DROP
User uploads the TikTok CSV harvest (comments + video data)
We remove the noise to focus on what matters.
3. THE SCRUB DOWN
System cleans noise & prepares text for analysis
AI analyzes the conversations to find meaning and patterns.
4. THE BRAIN SCAN
NLP extracts sentiment & key signals (keywords, topics)
Get clear, actionable insights in one dashboard.
5. THE INSIGHT REVEAL
Results hit the dashboard (sentiment, keywords, insights, recommendations)
Want to dig deeper or refine results?
6. NEED MORE DEPTH?
No
Re-filter data / adjust keywords
Yes
Export & use insights to grow your content with confidence.
7. FINAL ACTION
Export insights & improve content strategy

UX Design

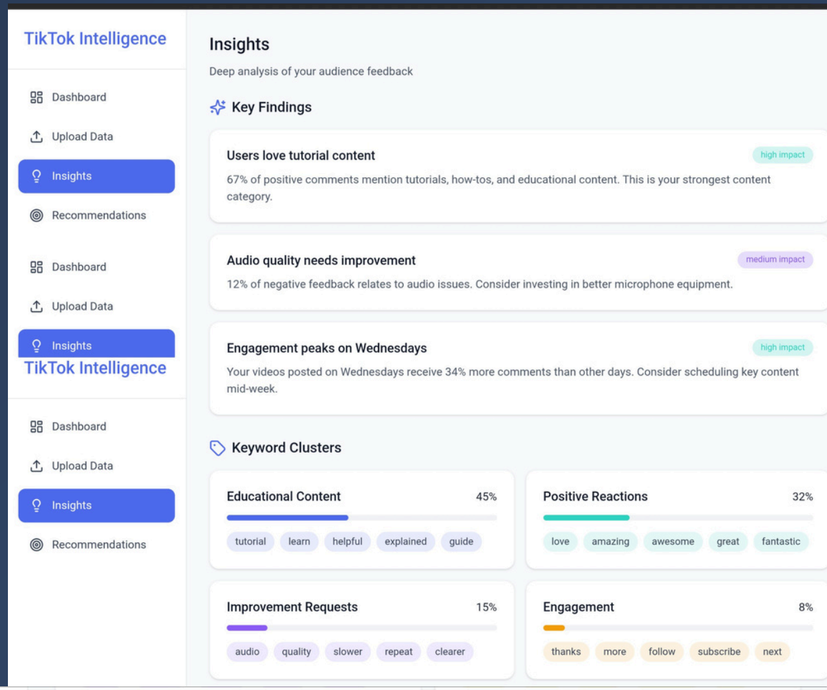

TikTok Intelligence
Dashboard
Upload Data
Insights
Recommendations
Dashboard
Overview of your TikTok comment analysis
Total Comments
1,247
% Positive
62%
% Negative
18%
% Neutral
20%
Sentiment Distribution
Positive 62%
Neutral 20%
Negative 18%
Top Keywords
240
180
120
60
0
tutorial
amazing
love
helpful
Keyword Frequency
tutorial
234
Example Comments
Positive
Quick Insights
Users love your tutorial content
Audio quality mentione of negative comments
Engagement peaks on

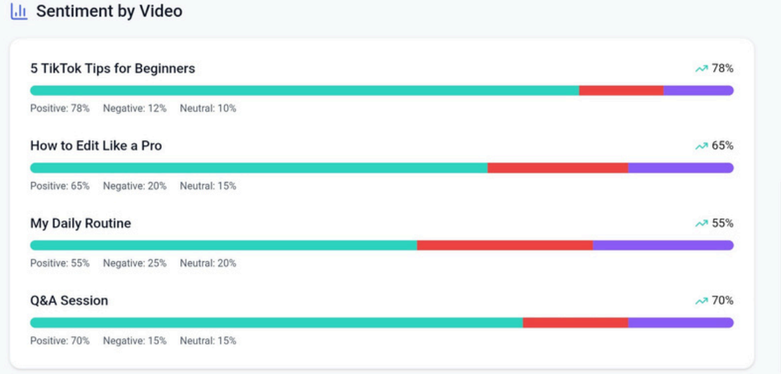
TikTok Intelligence
Dashboard
Upload Data
Insights
Recommendations
Upload Data
Upload your TikTok comments CSV file for analysis
Drag and drop your CSV file here
or click to browse
Select File
File Requirements
CSV format with columns: comment_text, timestamp, username
Maximum file size: 10MB
UTF-8 encoding recommended

Upload Data
Insights
Recommendations
Dashboard
Upload Data
Insights
Recommendations
Dashboard
Upload Data
Insights
Recommendations
Insights
Recommendations
Upload Data
Insights
Recommendations

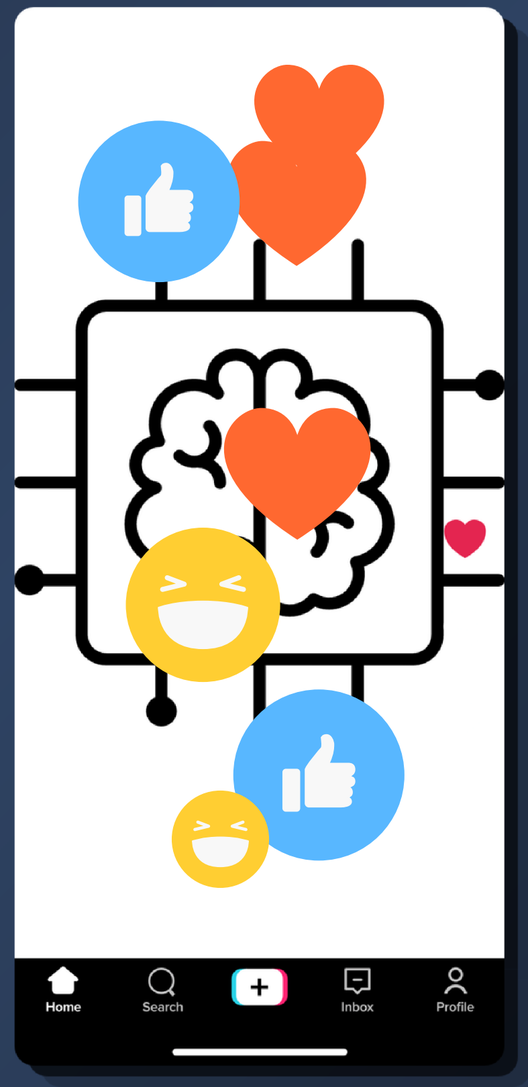

Tutorial-style content
67% of positive comments mention educational value
Create more step-by-step guides and how-to videos
Post on Wednesdays
34% higher engagement on mid-week content
Schedule your best content for Wednesday uploads
Respond to comments
Users appreciate creator interaction
Reply to at least top 20 comments within 24 hours
What to Improve
Audio quality
12% of negative feedback mentions audio issues
Invest in a better microphone or audio editing software
Video pacing
8% of viewers find content too fast
Slow down explanations or add on-screen text
Consistency
Viewers asking for regular uploads
Create a posting schedule and stick to it
Suggested Content Ideas
Advanced editing techniques
High confidence
Follows your successful tutorial pattern
tutorial
educational
trending
Behind the scenes of conten creation
Users are curious about your process
personal
engagement

TikTok Intelligence
Dashboard
Upload Data
Insights
Recommendations

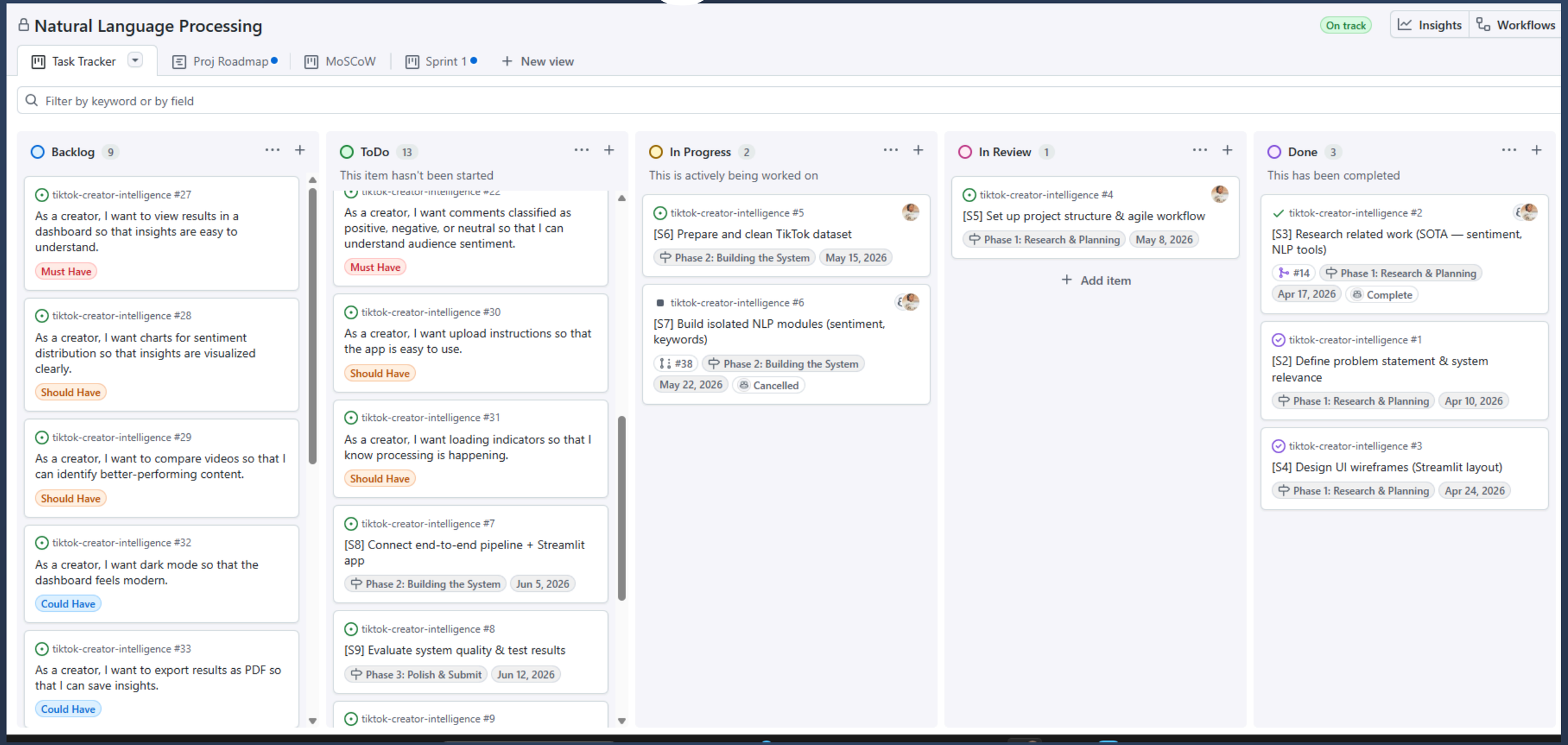

TikTok Intelligence
Dashboard
Upload Data
Insights
Recommendations
Dashboard
Upload Data
Insights
TikTok Intelligence
Dashboard
Upload Data
Insights
Recommendations
Insights
Deep analysis of your audience feedback
Key Findings
Users love tutorial content
high impact
67% of positive comments mention tutorials, how-tos, and educational content. This is your strongest content category.
Audio quality needs improvement
medium impact
12% of negative feedback relates to audio issues. Consider investing in better microphone equipment.
Engagement peaks on Wednesdays
high impact
Your videos posted on Wednesdays receive 34% more comments than other days. Consider scheduling key content mid-week.
Keyword Clusters
Educational Content
45%
tutorial
learn
helpful
explained
guide
Positive Reactions
32%
love
amazing
awesome
great
fantastic
Improvement Requests
15%
audio
quality
slower
repeat
clearer
Engagement
8%
thanks
more
follow
subscribe
next


Sentiment by Video
5 TikTok Tips for Beginners
78%
Positive: 78%
Negative: 12%
Neutral: 10%
How to Edit Like a Pro
65%
Positive: 65%
Negative: 20%
Neutral: 15%
My Daily Routine
55%
Positive: 55%
Negative: 25%
Neutral: 20%
Q&A Session
70%
Positive: 70%
Negative: 15%
Neutral: 15%

Deliverable 4
Agile Plan



Task 3 Agile Board



08/05/2026

Task 3 MoSCoW View

Must Have 7

tiktok-creator-intelligence #20
As a creator, I want to upload a CSV file so that I can analyze TikTok comments.
Backlog

tiktok-creator-intelligence #21
As a creator, I want the system to clean comment text so that the analysis is more accurate.
Backlog

tiktok-creator-intelligence #22
As a creator, I want comments classified as positive, negative, or neutral so that I can understand audience sentiment.
Backlog

tiktok-creator-intelligence #23
As a creator, I want to view sentiment statistics so that I can quickly understand audience reactions.
Backlog

tiktok-creator-intelligence #24
As a creator, I want the system to extract keywords so that I can identify common discussion topics.

Should Have 4

tiktok-creator-intelligence #28
As a creator, I want charts for sentiment distribution so that insights are visualized clearly.
Backlog

tiktok-creator-intelligence #29
As a creator, I want to compare videos so that I can identify better-performing content.
Backlog

tiktok-creator-intelligence #30
As a creator, I want upload instructions so that the app is easy to use.
Backlog

tiktok-creator-intelligence #31
As a creator, I want loading indicators so that I know processing is happening.
Backlog

+ Add item

Could Have 3

tiktok-creator-intelligence #32
As a creator, I want dark mode so that the dashboard feels modern.
Backlog

tiktok-creator-intelligence #33
As a creator, I want to export results as PDF so that I can save insights.
Backlog

tiktok-creator-intelligence #34
As a creator, I want keyword word clouds so that topics are easier to explore.
Backlog

Won't Have 3

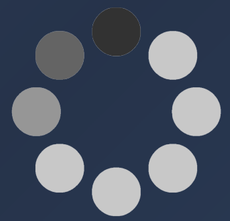
tiktok-creator-intelligence #35
As a creator, I want real-time TikTok API integration.
Backlog

tiktok-creator-intelligence #36
As a creator, I want mobile app support.
Backlog

tiktok-creator-intelligence #37
As a creator, I want multilingual support.
Backlog

08/05/2026

Task 3 Sprint View



Sprint 1 4 ··· +

May 08 - May 21

tiktok-creator-intelligence #20

As a creator, I want to upload a CSV file so that I can analyze TikTok comments.

ToDo

tiktok-creator-intelligence #21

As a creator, I want the system to clean comment text so that the analysis is more accurate.

ToDo

tiktok-creator-intelligence #30

As a creator, I want upload instructions so that the app is easy to use.

ToDo

tiktok-creator-intelligence #31

As a creator, I want loading indicators so that I know processing is happening.

ToDo

Sprint 2 0 Current ··· +

May 22 - Jun 04

Sprint 3 0 ··· +

Jun 05 - Jun 18

+ Add item

08/05/2026

Acceptance Criteria

As a creator, I want the system to extract keywords so that I can identify common discussion topics. #24

Open · Chinel0/tiktok-creator-intelligence Private

Chinel0 opened 2 weeks ago · Last edited by Chinel0

Acceptance Criteria:

System identifies the most frequently used words (topics).

System filters out "stop words" (e.g., "and", "the", "a").

A list of the top 5-10 keywords is displayed to the user.

Create sub-issue

Chinel0 moved this to Backlog in Natural Language Processing 2 weeks ago

Chinel0 added this to Natural Language Processing 2 weeks ago

Chinel0 moved this from Backlog to ToDo in Natural Language Processing 2 weeks ago

Assignees: No one - Assign yourself · Assign to Agent

Labels: No labels

Projects

As a creator, I want to upload a CSV file so that I can analyze TikTok comments. #20

Edit

Open · Chinel0/tiktok-creator-intelligence Private

Chinel0 opened 2 weeks ago · Last edited by Chinel0

Acceptance Criteria:

User can select a .csv file from their computer.

The system checks if the file format is correct.

Data from the CSV appears in the app after upload.

Create sub-issue

Chinel0 moved this to Backlog in Natural Language Processing 2 weeks ago

Chinel0 added this to Natural Language Processing 2 weeks ago

Chinel0 moved this from Backlog to ToDo in Natural Language Processing 2 weeks ago

Add a comment

Assignees: No one - Assign yourself · Assign to Agent

Labels: No labels

Projects: Natural Language Processing

Status: ToDo

dates: No date

MoSCoW: Must Have

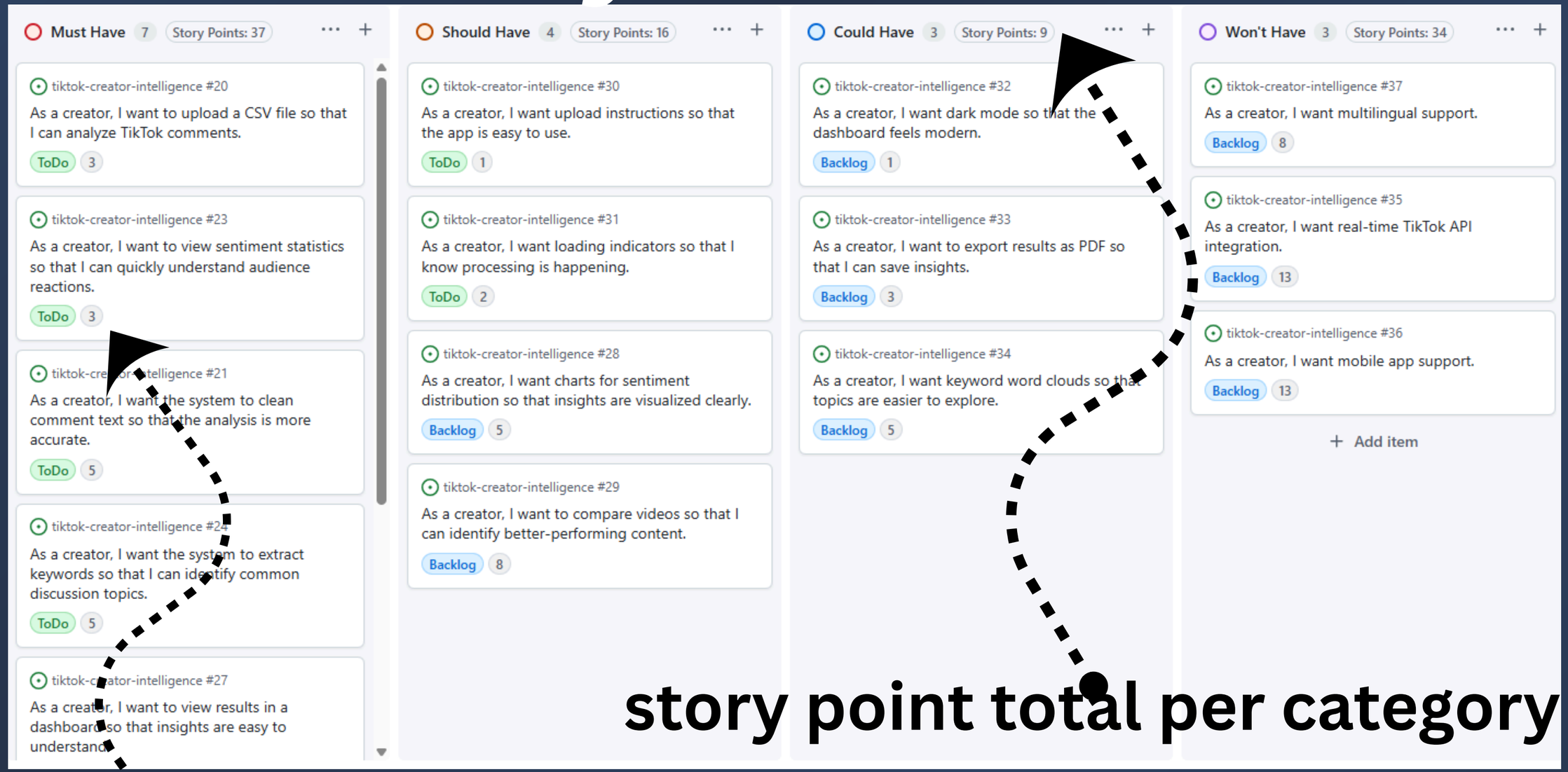
Story Points: 3

Sprint: Sprint 1 • May 8 - May 21

Task 3

08/05/2026

Task 3 Story Points

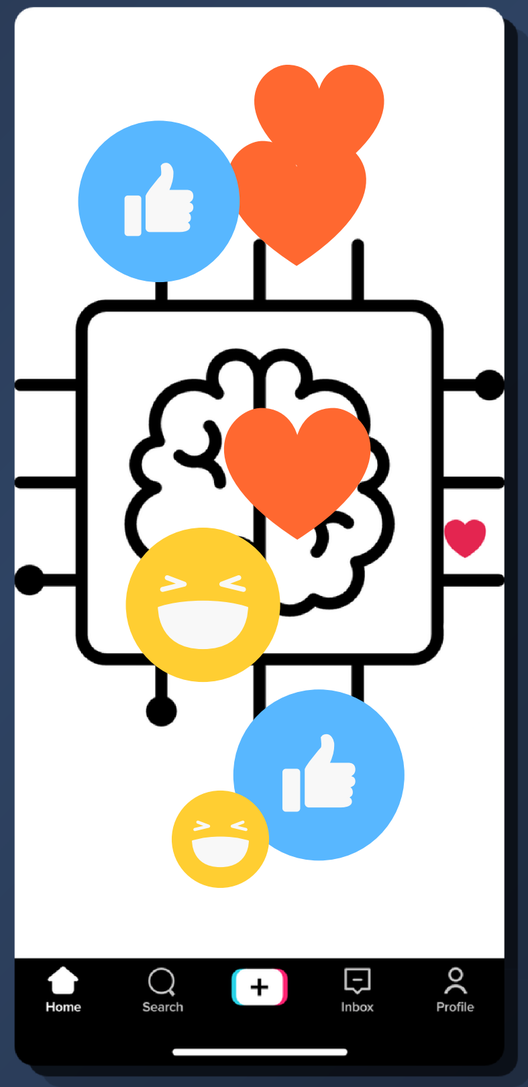


story point total per category

Story point per task

08/05/2026

Deliverable 5
Data Strategy



Data Source

Primary Data Source
Personal TikTok account — @ichbinnelo (public profile)

Retrieval Method
Custom Python scraper using Playwright browser automation. Intercepts TikTok's internal comments API (/api/comment/list/) directly from the network — no third-party library.

Data Scope

- 161 videos
- 1,211 comments
- 9 content categories
- scraped May 2026

15/05/2026

Data Lineage & Legal Check

✅ **Copyright**	my own account no third party rights
✅ **Terms of Use**	Personal data collection for educational research
✅ **GDPR**	Public comments, no PII stored, university project only
✅ **Scientific**	Full citation: Ariyus, D., Manongga, D., & Sembiring, I. (2024). Enhancing Sentiment Analysis of Indonesian Tourism Video Content Commentary on TikTok: A FastText and Bi-LSTM Approach. Engineering, Technology & Applied Science Research, 14(6), 18020–18028.

15/05/2026

Pre-processing

```python
import re

def clean_text(text):
    if not isinstance(text, str): return ""
    text = text.lower()
    text = re.sub(r'http\S+|www\S+', '', text)
    text = re.sub(r'@\w+',           '', text)
    text = re.sub(r'#\w+',           '', text)
    text = re.sub(r'[^\w\s]',        '', text)
    text = re.sub(r'\d+',            '', text)
    text = re.sub(r'[^\x00-\x7F]+',  '', text)
    text = re.sub(r'\s+',            ' ', text)
    tokens = [t for t in text.split() if len(t) >= 3]
    return ' '.join(tokens)

examples = [
    "omg I love this so much!!! 😍😍 #foodtok #fyp check https://t.co/abc123",
    "@ichbinnelo please do more cooking videos!! 😻",
    "this and starting my seedlings #CapCut #foodtok",
    "I wish you a wonderful day ❤️",
    "berlin is nice for tourists but not for residents",
]

print(f'{"RAW INPUT":<55}  →  CLEANED OUTPUT')
print('-' * 95)
for raw in examples:
    cleaned = clean_text(raw)
    print(f'{raw[:55]:<55}  →  {cleaned if cleaned else "(empty after cleaning)"}')
```

Python

```
RAW INPUT                                                →  CLEANED OUTPUT
-----------------------------------------------------------------------------------------------
omg I love this so much!!! 😍😍 #foodtok #fyp check https  →  omg love this much check
@ichbinnelo please do more cooking videos!! 😻            →  please more cooking videos
this and starting my seedlings #CapCut #foodtok          →  this and starting seedlings
I wish you a wonderful day ❤️                             →  wish you wonderful day
berlin is nice for tourists but not for residents        →  berlin nice for tourists but not for residents
```

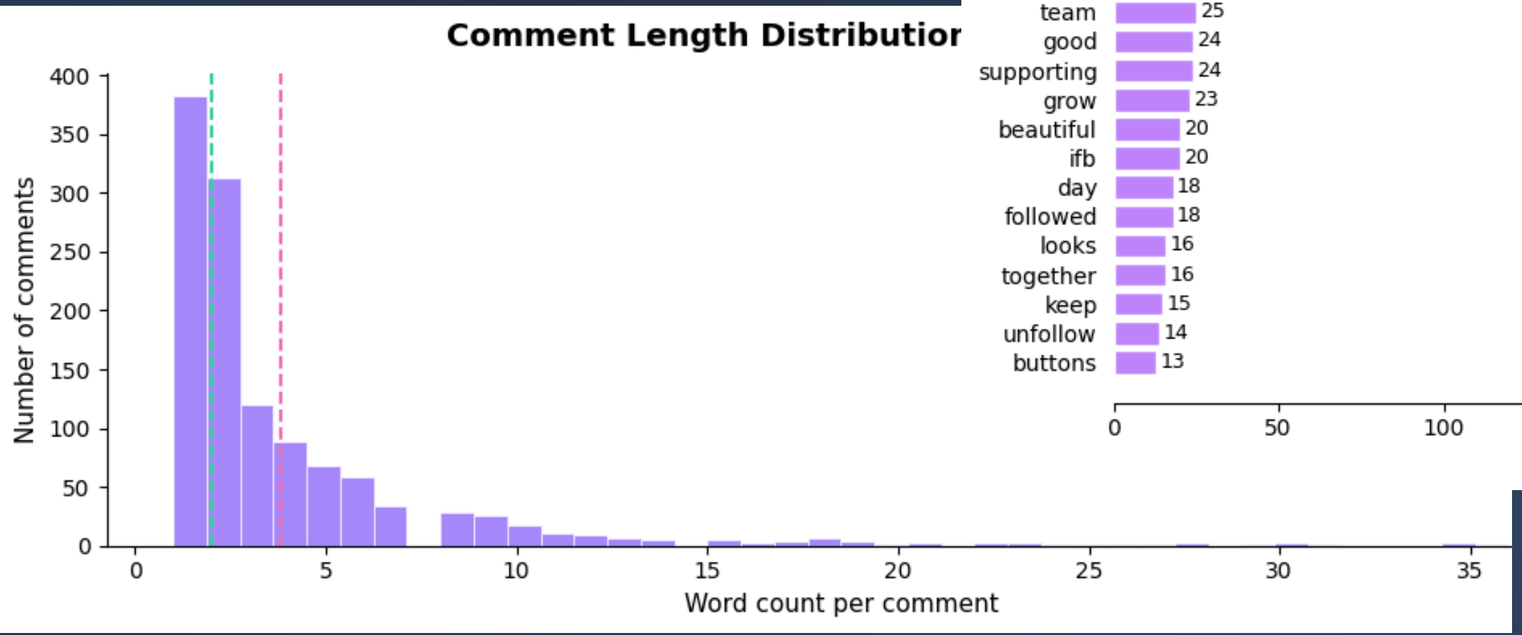
15/05/2026

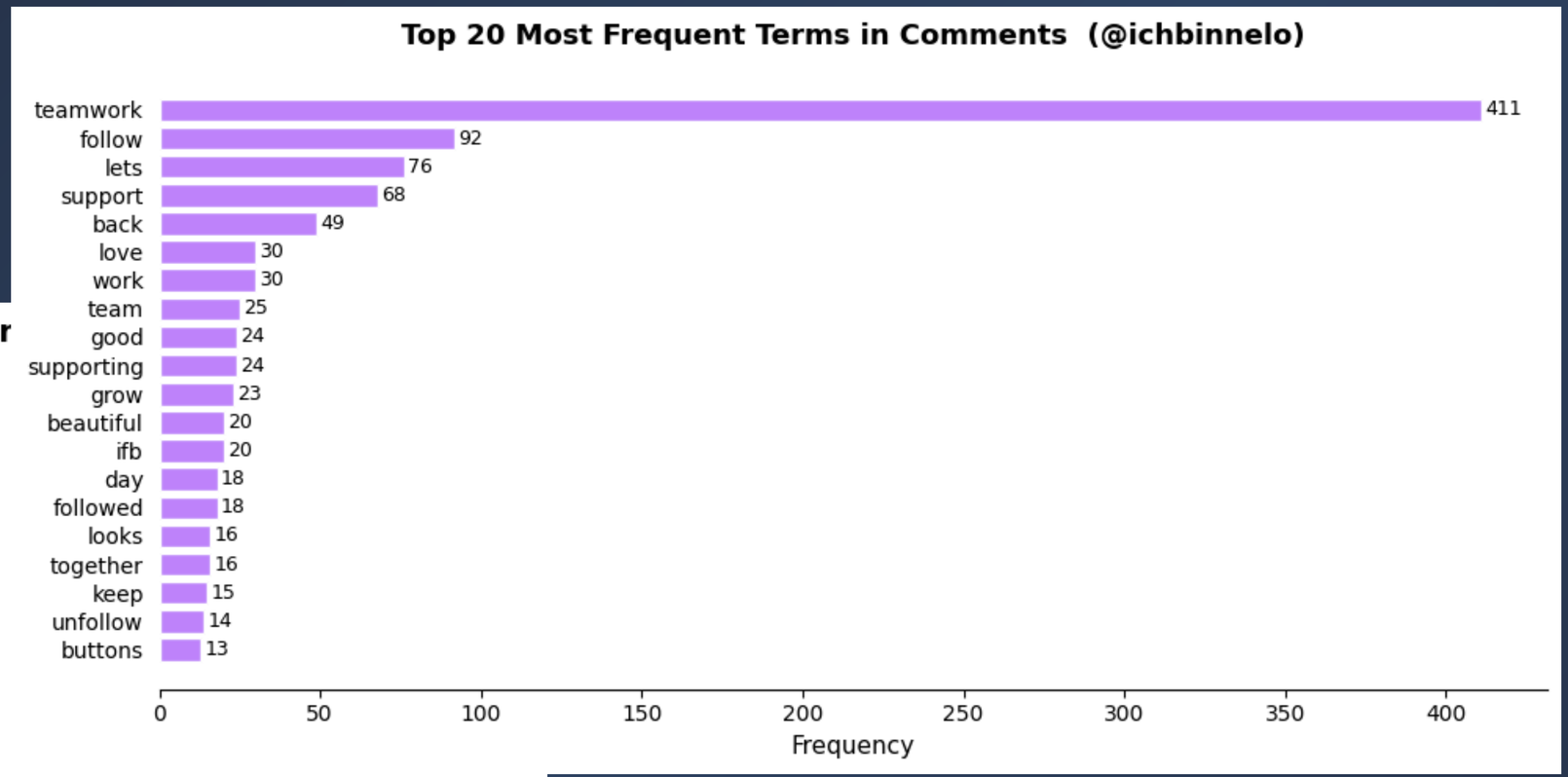
Exploratory Showcase

Total comments: 1,211

English comments: 944 (78%) — remaining 267 were auto-translated from German and other languages

Average comment length: 3.8 words — TikTok comments are very short, which is why VADER was chosen over longer-context models





15/05/2026

Applicability

SIGNAL → comment_text (what the audience said)

SIGNAL → like_count (how many agreed with the comment)

SIGNAL → comment_date (spot trends over time)

SIGNAL → video_id (links comment to content category)

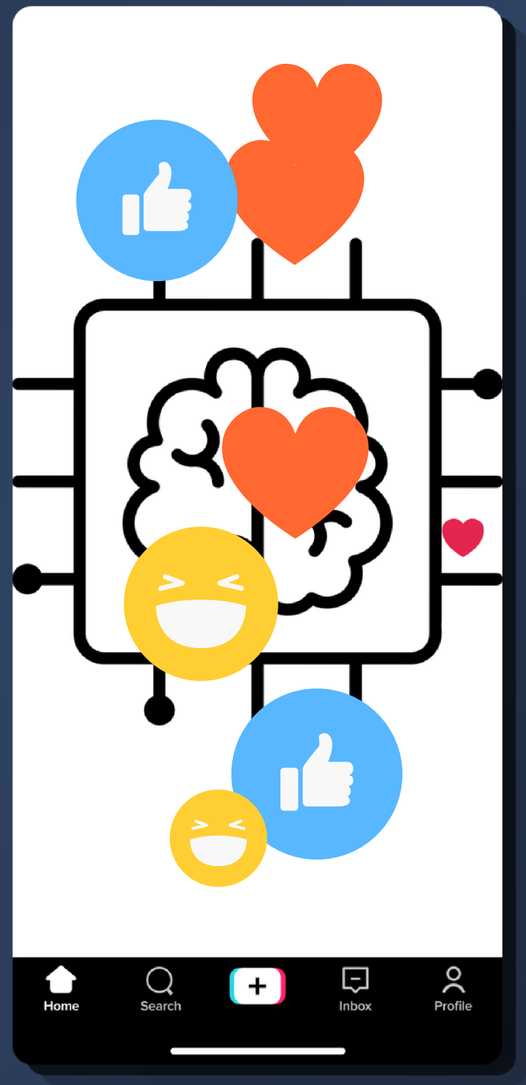
NOISE → scraped_at (internal timestamp, not analytical)

NOISE → original_text (pre-translation backup, not used

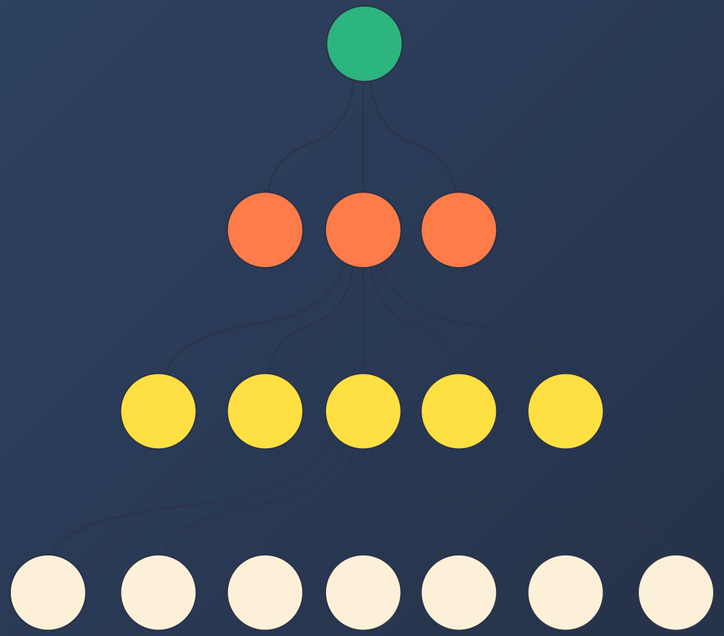
NOISE → language (processing artifact, job done)

15/05/2026

Deliverable 6
NLP Modeling



Method & Model Choice



Method 1 — Sentiment Classification
Classify each comment as Positive, Negative, or Neutral based on the words used.

Model: VADER — vaderSentiment v3.3.2 (Hutto & Gilbert, 2014)

Why: Built specifically for short informal social media text. Handles slang, caps ("LOVE THIS"), exclamation marks, and emojis. Free, offline, no training data needed. Scores 1,211 comments in under 1 second.

Method 2 — Keyword Extraction
Find the most important words per content category to show what each audience talks about.

Model: TF-IDF — scikit-learn (unigrams + bigrams, English stopwords removed)

Why: Identifies words that are common in one category but rare in others — exactly the "what do food viewers say that lifestyle viewers don't?" question. Fully interpretable, no GPU, no training data.

Technical Setup

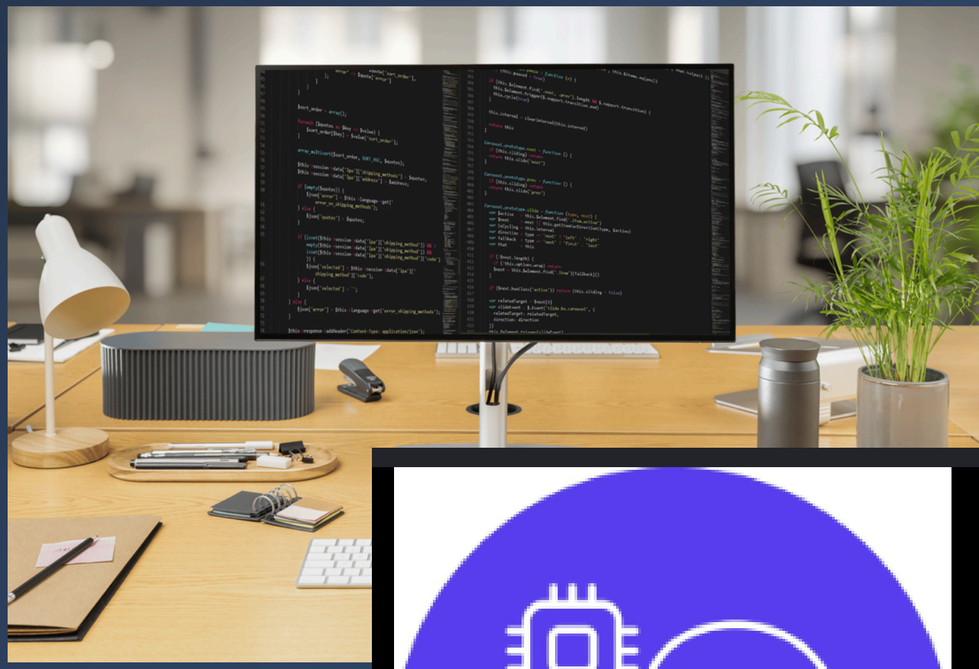
Environment: Local machine · Windows 11 · Python 3.12 · Jupyter Notebook

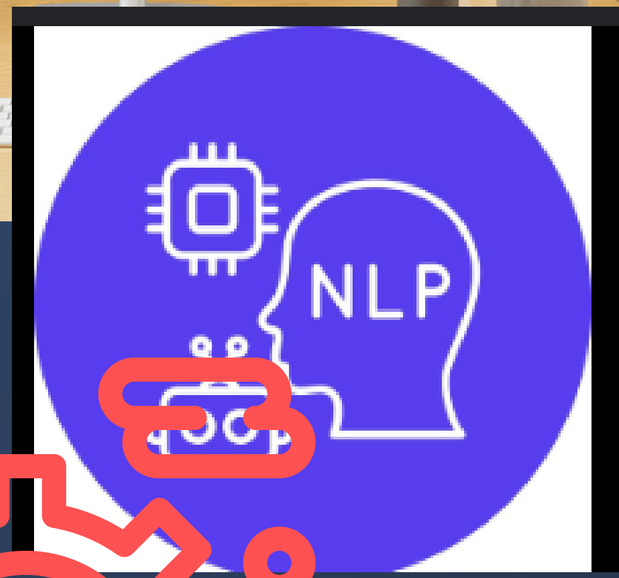
Library Role

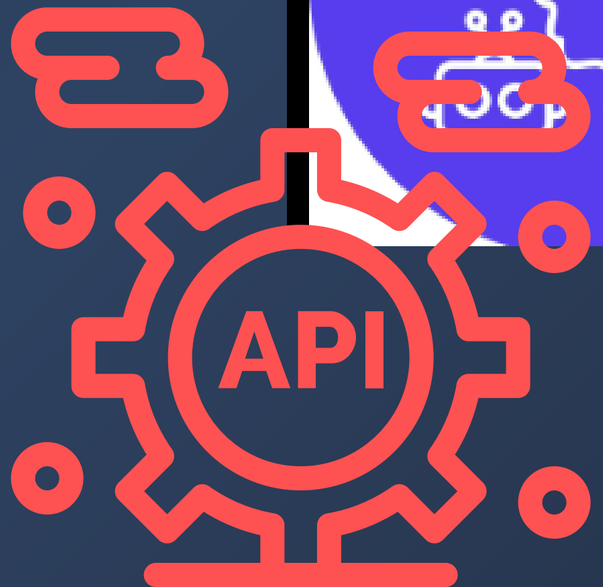
- vaderSentiment Sentiment scoring
- scikit-learn TF-IDF keyword extraction
- pandas Data loading and filtering
- nltk Stopword list
- deep-translator Pre-processing: German → English

Access: No API key. No account. No internet after install. Fully private, comment data never leaves the machine.

Cost & Latency: £0.00 · Full pipeline runs in under 5 seconds







Minimal Working Example

```python
# ── Minimal Working Example: Sentiment Classification ──────────────────
# Input: raw comment text  →  Output: compound score + label
# Model: VADER (vaderSentiment, Hutto & Gilbert 2014)

from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
import pandas as pd

analyzer = SentimentIntensityAnalyzer()

def score(text):
    c = analyzer.polarity_scores(str(text))['compound']
    label = 'POSITIVE' if c >= 0.05 else ('NEGATIVE' if c <= -0.05 else 'NEUTRAL')
    return round(c, 3), label

# Run on 5 real comments from the dataset
comments = pd.read_csv('data/comments_20260526_233400.csv', encoding='utf-8-sig')
sample   = comments['comment_text'].dropna().sample(5, random_state=42).tolist()

print(f'{"COMMENT":<50}  {"SCORE":>7}  LABEL')
print('─' * 75)
for text in sample:
    compound, label = score(text)
    print(f'{str(text)[:50]:<50}  {compound:>+.3f}  {label}')
```

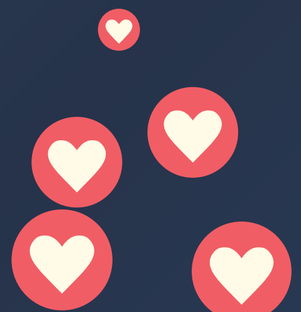
```
COMMENT                                             SCORE  LABEL
───────────────────────────────────────────────────────────────────────────
Wow yummy 😋 with my favorite song 🥹                +0.881  POSITIVE
let's get this started 👍                           +0.000  NEUTRAL
Another day another lets go lets roll 😄👍🏻✨        +0.778  POSITIVE
nice routinee                                       +0.421  POSITIVE
Won't unfollow                                      +0.000  NEUTRAL
```

```python
videos   = pd.read_csv('data/videos_merged.csv',              encoding='utf-8-sig')

# Join so each comment inherits its video's category
merged = comments.merge(videos[['video_id','video_type']], on='video_id', how='left')

def clean(text):
    text = re.sub(r'[^a-zA-Z\s]', ' ', str(text).lower())
    return re.sub(r'\s+', ' ', text).strip()

# Extract keywords for one category — change this to explore others
CATEGORY = 'Food & Cooking'
corpus   = merged[merged['video_type'] == CATEGORY]['comment_text'].dropna().apply(clean).tolist()

vectorizer = TfidfVectorizer(max_features=200, stop_words='english',
                             ngram_range=(1, 2), min_df=2)
vectorizer.fit(corpus)

scores   = np.array(vectorizer.transform(corpus).mean(axis=0)).flatten()
top_idx  = scores.argsort()[::-1][:10]
keywords = [(vectorizer.get_feature_names_out()[i], round(scores[i], 4)) for i in top_idx]

print(f'Top 10 keywords — {CATEGORY}')
print('─' * 35)
for word, score in keywords:
    bar = '' * int(score * 300)
    print(f'  {word:<22} {score:.4f}  {bar}')
```

```
Top 10 keywords — Food & Cooking
─────────────────────────────────
  yummy                  0.0966  ████████████████████████████
  looks                  0.0917  ███████████████████████████
  looks yummy            0.0455  █████████████
  thanks                 0.0416  ████████████
  think                  0.0416  ████████████
  just                   0.0416  ████████████
  love                   0.0387  ███████████
```

Sample Output & Observations

```
COMMENT                                                        SCORE  LABEL
--------------------------------------------------------------------------------
omg I love this so much it looks amazing                      +0.840  POSITIVE
this is my favorite type of content please keep making        +0.649  POSITIVE
I wish you a wonderful day                                    +0.751  POSITIVE
berlin is nice for tourists but not for residents             +0.226  POSITIVE
what was the first thing in the video this one with str       +0.572  POSITIVE
it is not such a bad winter day is it                         +0.431  POSITIVE
please do it because I am not that famous yet                 +0.318  POSITIVE
I have been trying to pay off my debt for years this re       -0.440  NEGATIVE
I add back everyone who adds me                               +0.000  NEUTRAL
My condolences                                                +0.000  NEUTRAL

Full corpus sentiment distribution:
  POSITIVE     723  (59.7%)  ########################
  NEUTRAL      433  (35.8%)  ##############
  NEGATIVE      55  ( 4.5%)  #
```



Observations:

- Genuine compliments and encouragement are scored confidently and correctly
- Negative sentiment is rare in the corpus — expected for a creator's own comment section
- Very short comments ("wow", "same", "done") score neutral — not enough words for VADER to read
- Engagement-bait comments ("teamwork lets go") score weakly positive despite carrying no real opinion about the content — the video_type filter removes these before analysis

Critical Scaling Points

1. VADER misses sarcasm
"sure, love spending all my money on debt 🙃" scores positive. Fine at 1,211 comments. At 50,000 it misleads the dashboard.

Mitigation: Flag scores between -0.2 and +0.2 as uncertain rather than neutral.

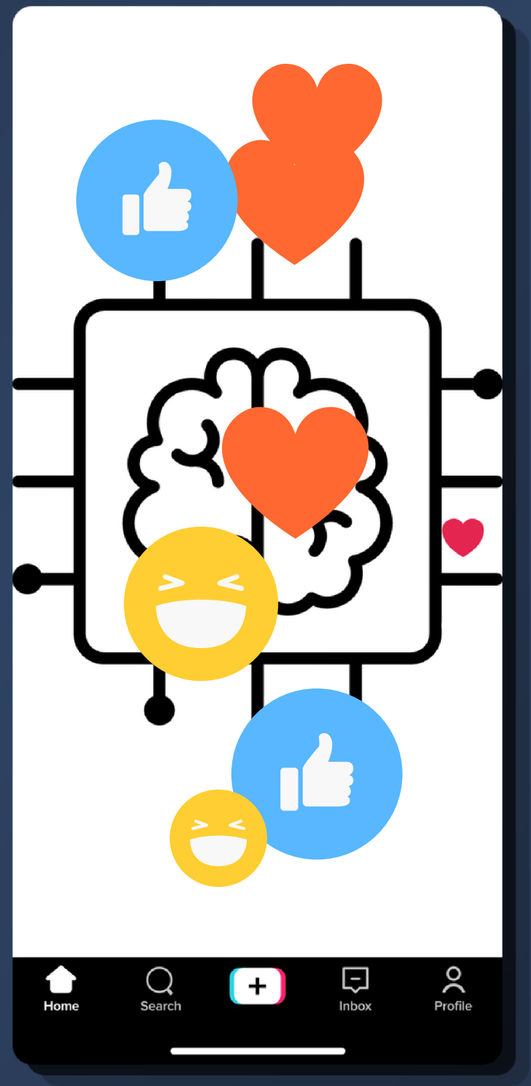
2. TF-IDF breaks on small categories
Books & Reading has only 4 videos — too few documents for reliable keywords.
Mitigation: Suppress keyword output for categories with fewer than 10 videos.

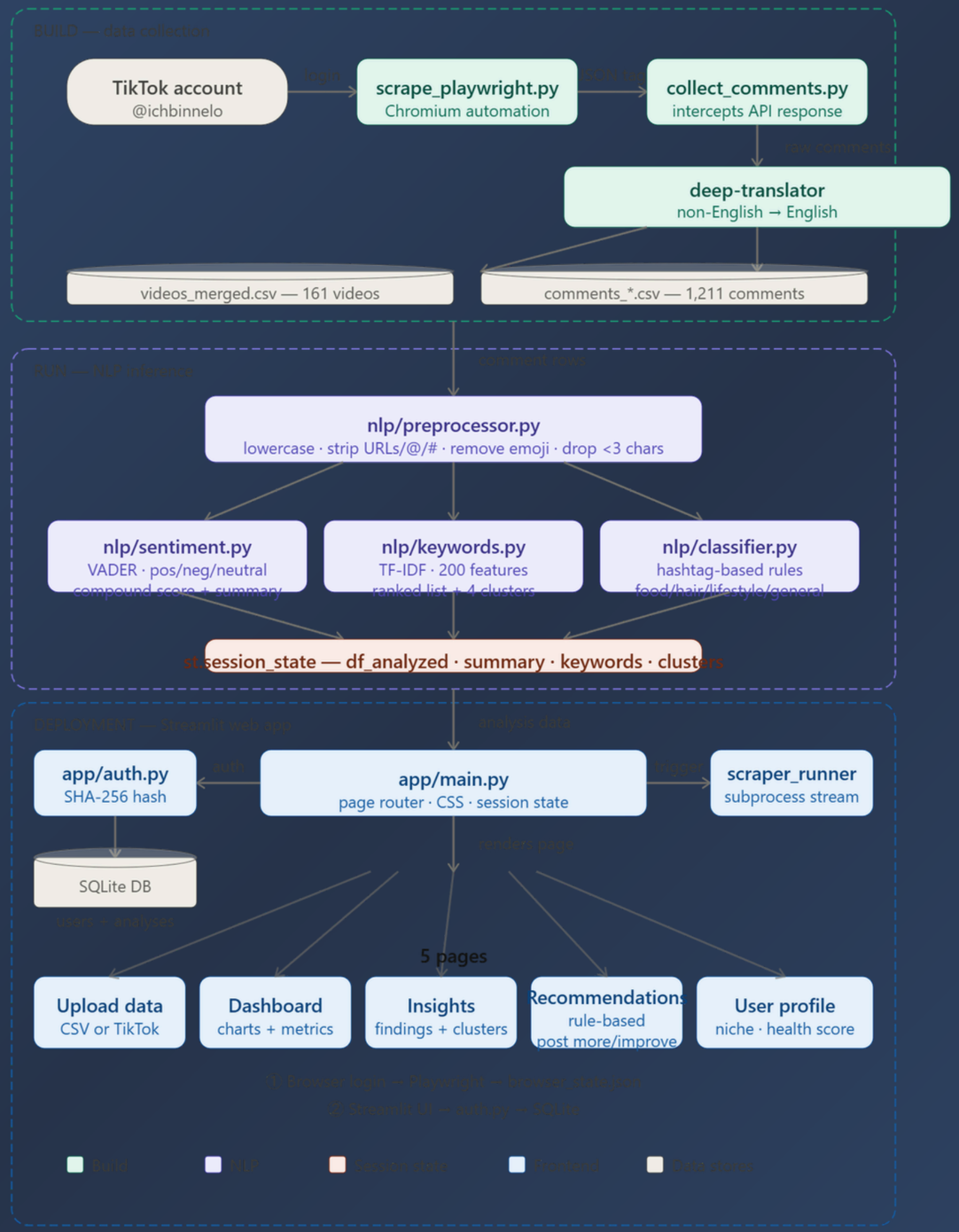
3. Translation rate limits
deep-translator (Google Translate) has undocumented free-tier limits. At 10,000+ comments it would fail mid-run.
Mitigation: Batch in groups of 100 with a short delay, or switch to offline Helsinki-NLP/opus-mt.

4. Video categories don't generalise
The hashtag rules were written for @ichbinnelo. A different creator would get mostly Unknown.
Mitigation: Make category rules a user-configurable file in the app.

Deliverable 7

End-2-End-Systems



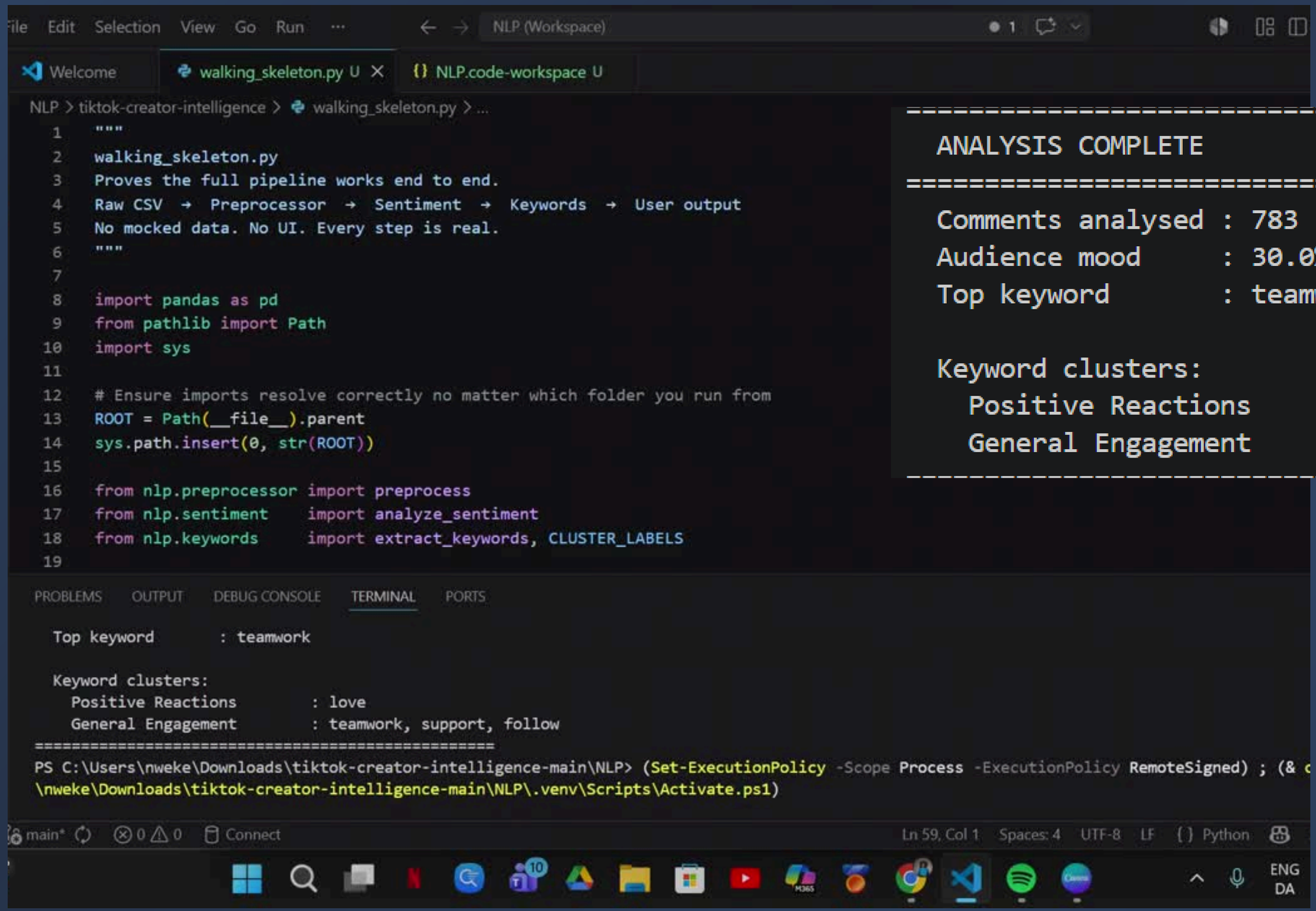


System Architecture Diagram

Pipeline Component Breakdown

Junction	Input Format	Tool	Output Format
Raw Data → Preprocessor	Excel file (.xlsx), 1211 rows, 15 columns	pandas + preprocessor.py	Cleaned DataFrame, 944 English rows, new clean_comment column
Preprocessor → Sentiment	Cleaned DataFrame with clean_comment column	VADER (nltk) via sentiment.py	DataFrame + new sentiment column (positive/negative/neutral) + score
Preprocessor → Keywords	Cleaned DataFrame with clean_comment column	TF-IDF (scikit-learn) via keywords.py	List of top 20 keywords + 4 keyword clusters with percentages
NLP Modules → Insights	Sentiment labels + keyword clusters	insights.py	Dictionary of findings and recommendations
Insights → UI	Python dictionaries and DataFrames	Streamlit via app/main.py	Visual dashboard with charts, cards, recommendations
UI → Database	User profile + analysis results	SQLite via sqlite3	Stored analysis history for returning users

Walking Skeleton Demo




```
==============================================
  ANALYSIS COMPLETE
==============================================
  Comments analysed : 783
  Audience mood     : 30.0% positive
  Top keyword       : teamwork

  Keyword clusters:
    Positive Reactions        : love
    General Engagement        : teamwork, support, follow
----------------------------------------------
```

Bottleneck Analysis

Bottleneck 1
35% of comments silently dropped 1,211 comments loaded → only 783 analysed. The preprocessor discards translated comments because their language column still says 'de' or 'fr' even though the text is already English.

Bottleneck 2
Keyword output polluted by engagement spam Top keyword is teamwork (442x). This comes from follow-train posts, not real content feedback. TF-IDF runs on all comments at once

Next 3 Task
1. Fix the language filter — keep translated comments that have an original_text value. Recovers the missing 428 comments.

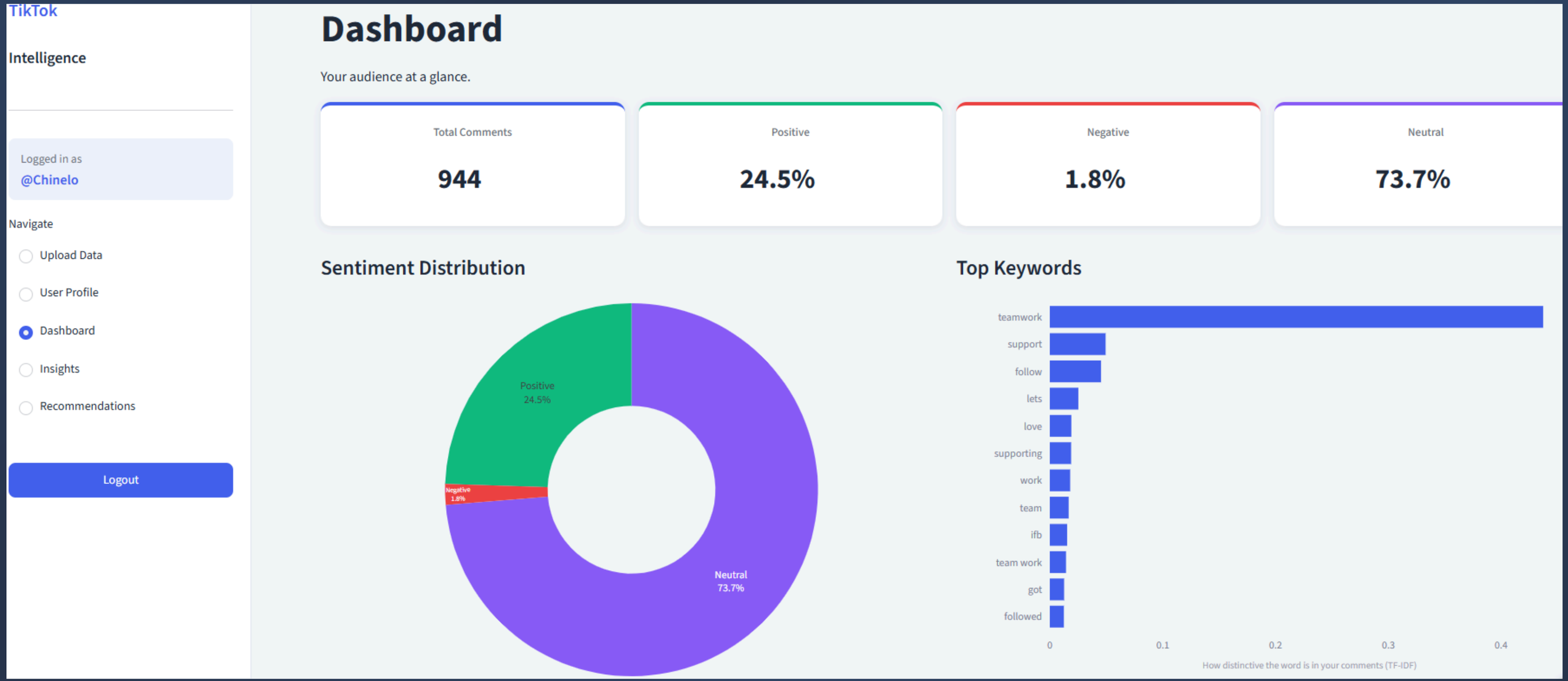
2. Filter out Engagement / Growth videos before keyword extraction. Removes the teamwork/follow spam and surfaces real content keywords.

3. Connect pipeline output to the Streamlit dashboard — replace print() statements with live UI output. This completes the bridge from backend to prototype.

What's Under the Hood?

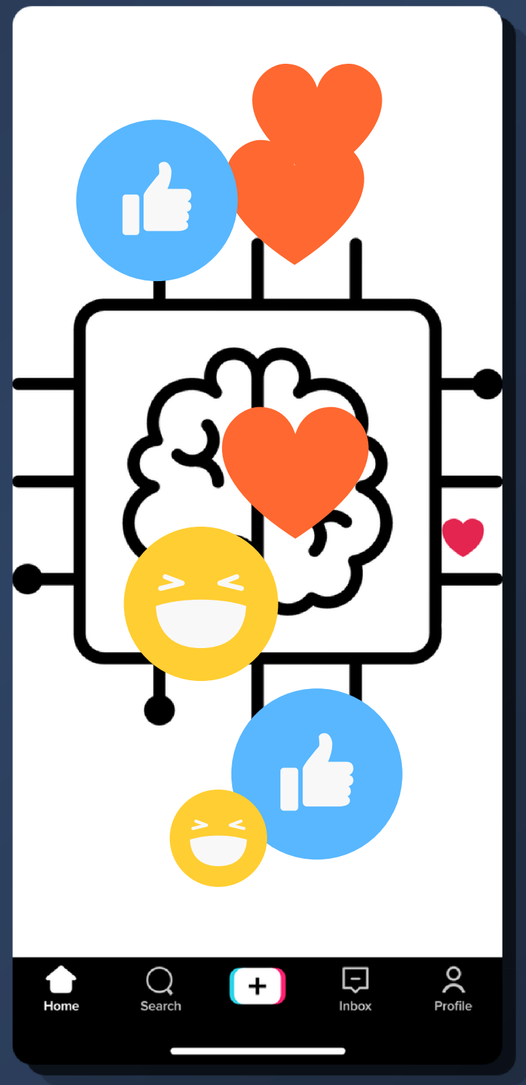
NLP Model Used
VADER: open-source sentiment lexicon

TF-IDF: a statistical method, plus rule-based analytics



Deliverable 8

Evaluation & Quality



Performance Metrics

Metric	Score	Justification
Accuracy	91.90%	Overall correctness; test set mirrors the real class balance
Macro F1	94.10%	Weighs all 3 classes equally — the rare negative class matters most to creators
Negative-class F1	100%	The high-value class: criticism detection (2/2 caught)

Gain over naive baseline:

+37.8 pts accuracy

+70.7 pts Macro F1

Key Finding from Evaluation

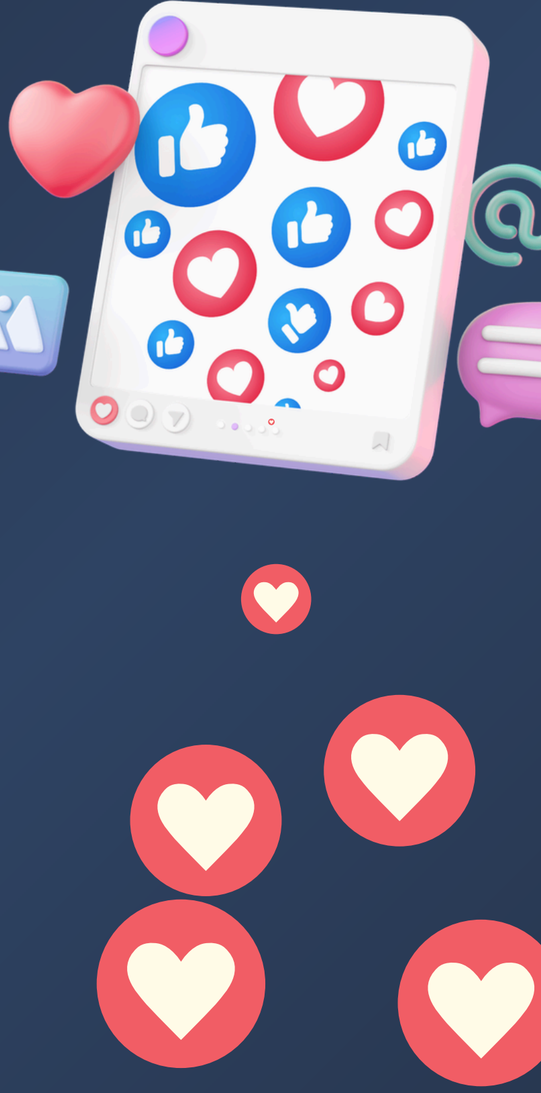
Evaluated on 37 real comments from my own account, manually labeled (held-out test set). Preprocessing stripped the emojis and punctuation VADER uses as sentiment signals.

Fix applied: VADER now runs on the original comment text; cleaning is kept for TF-IDF keywords only.

Result: +5.4 pts accuracy, +9.8 pts Macro F1 from the emoji fix.

Benchmark Comparison

System	Accuracy	Macro F1
Baseline A: Majority class (always positive)	54.10%	23.40%
Baseline B: VADER on cleaned text (emojis stripped)	86.50%	84.30%
Full system: VADER emoji-aware (current)	91.90%	94.10%



Keeping emojis through to the sentiment analyser contributes +5.4 pts accuracy and +9.8 pts Macro F1 over the ablated version, confirming that emoji-aware sentiment is the highest-value design decision in the pipeline. On TikTok, the emoji often is the sentiment.

Pipeline Efficiency

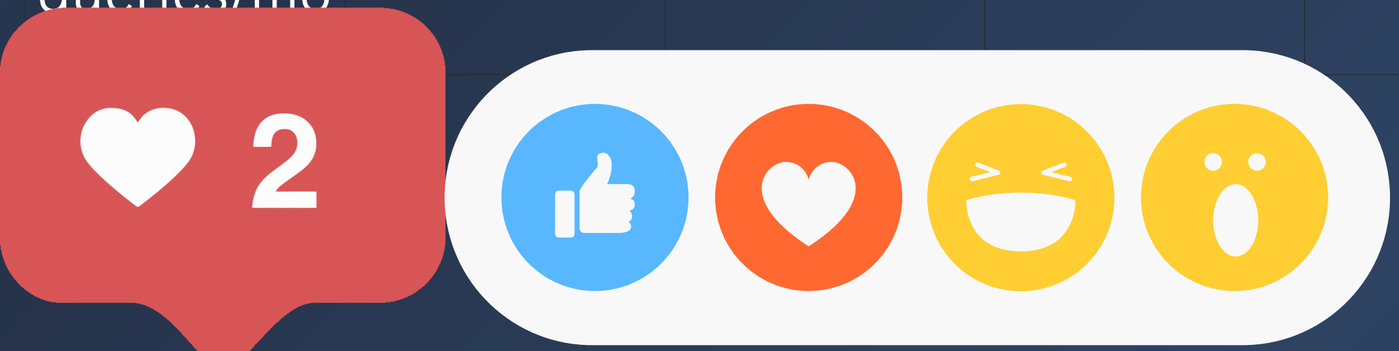
Scale	Avg Latency	Worst Case	Cost/Query	Est. Monthly
100 queries/mo	0.28 s	0.34 s	€0.00	€0.00
1,000 queries/mo	0.28 s	0.34 s	€0.00	€0.00
10,000 queries/mo	0.28 s	0.34 s	€0.00	€0.00

Bottleneck: sentiment scoring (VADER) 39% of total latency. Preprocessing is now only 9%.

Throughput: 4,300 comments/second end-to-end (1,211 comments + 161 videos in 0.28 s).

Cost: zero. Fully local pipeline, no API calls ,VADER + TF-IDF + statistics run on device.

Note: TikTok scraping (55 min for 161 videos) is a one-time data collection step, separate from the analysis pipeline.



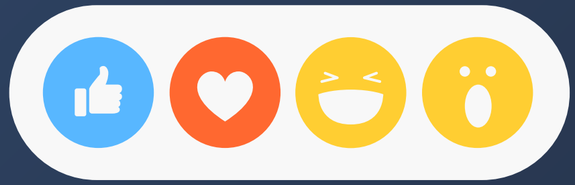
Error Analysis

Category	Count	% of data	Fix
Non-English comments excluded	267	22.00%	HIGH — trust short ASCII comments as English
language-detection false positives	142	11.70%	(1–2 word comments like "wow" misdetected)
Emoji-only comments	147	12.10%	**Fixed** — VADER now reads original text
Test-set misclassifications	3 of 37	8.10%	neutral/positive boundary, ±0.05 threshold
Potential sarcasm	5	0.40%	Route to transformer model (future work)

[DONE] Emoji sentiment VADER now runs on original text (+5.4 pts accuracy, measured)

[NEXT] Language filter trust short ASCII comments as English instead of langdetect's guess; recovers ~142 comments at zero risk

[FUTURE] Slang & sarcasm custom VADER lexicon for hype slang ("hard", "crazy", "sick"), or swap to cardiffnlp/twitter-roberta-base-sentiment



SUS Results per Question (n = 6)
Combined SUS score: 87.5 / 100

#	Statement	Avg (1–5)	Good =
1	I think that I would like to use this system frequently.	4.3	high ✓
2	I found the system unnecessarily complex.	1.7	low ✓
3	I thought the system was easy to use.	4.7	high ✓
4	I think that I would need the support of a technical person to be able to use this system.	1.3	low ✓
5	I found the various functions in this system were well integrated.	4.3	high ✓
6	I thought there was too much inconsistency in this system.	2	low ✓
7	I would imagine that most people would learn to use this system very quickly.	4.5	high ✓
8	I found the system very cumbersome to use.	1.3	low ✓
9	I felt very confident using the system.	4.8	high ✓
10	I needed to learn a lot of things before I could get going with this system.	1.3	low ✓

User Evaluation

Was anything flat-out WRONG with the app? What?

5 responses

Its a very good idea but the categorization of comments is very unreliable.

I didn't notice anything wrong, it looks okay to me and it's something I could use

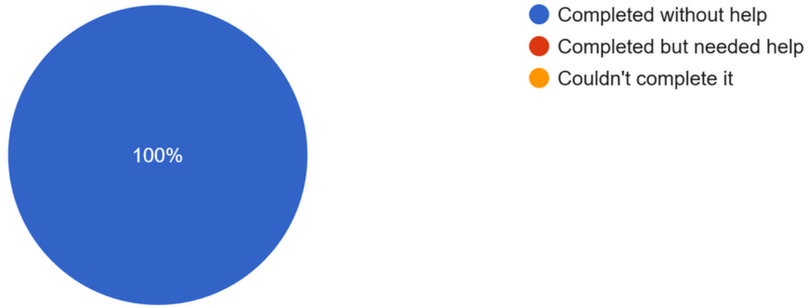
Some words like content and love content were flagged differently and that is confused as a user.

Slang being seen as negative

no because it tells me what to focus on when making contents for my audience, but I noticed some repetition of words, I feel like it still needs to be More specific and clear in breaking down the analysis of contents so as not to confuse new users.

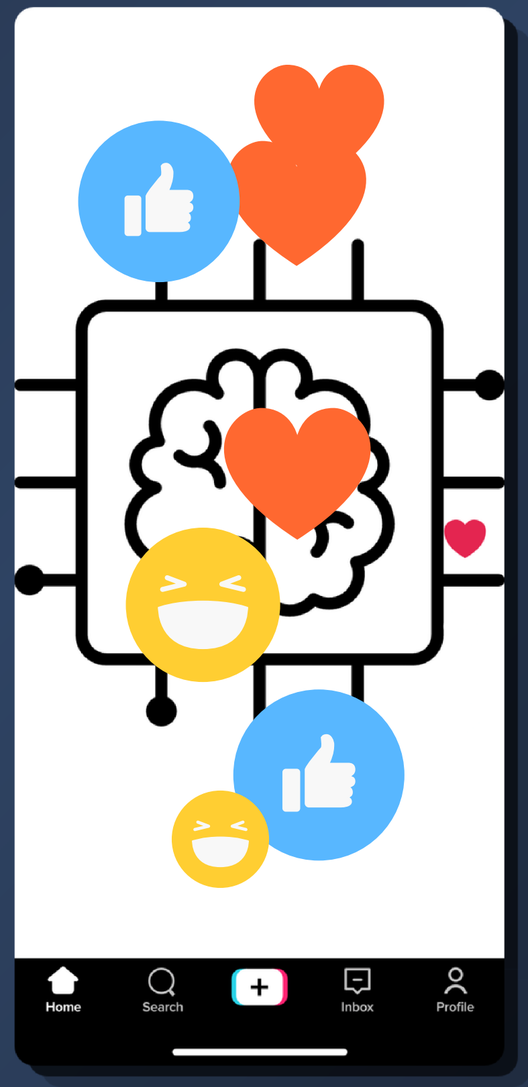
Register + upload both files + run analysis, how did it go?

5 responses



Deliverable 9

Optimizing your Prototype



Optimization Backlog

Natural Language Processing

On track | Insights | Workflows 7

Task Tracker | Proj Roadmap | MoSCoW | Sprint 1 | Sprint 2 | sprint3 | + New view

Filter by keyword or by field | View

	Title	Impact	Eff...	Priority	Acceptance Criterion
	Sprint 4 7 Jun 19 - Jun 28 Current				
24	[P1] Fix langdetect false positives for short ASCII-only comments #58	HIGH	S	P1	Non-English exclusion drops from 22.0% to 12.0% (142 short comments recovered)
25	[P1] Add custom VADER lexicon entries for TikTok hype slang #59	HIGH	S	P1	Macro F1 stays >= 94.1%; hype-slang words (hard/crazy/sick/dead) score positive o...
26	[P1] Automate regression test via test_harness.py on every git push (GitHub Action... #60	HIGH	S	P1	CI runs test_harness.py on every push; exits 1 if any quality flag fires
27	[P2] Show keyword cluster match reason in Dashboard #61	MEDIUM	M	P2	Each keyword cluster shows the seed token that triggered the assignment
28	[P2] Expand language support beyond German (Spanish, French, Portuguese) #62	HIGH	L	P2	Spanish, French, Portuguese translated and processed; total exclusion rate below 12%
29	[P2] Cache analysis result per file hash to skip re-running pipeline on same upload #63	MEDIUM	S	P2	Same file re-upload completes in under 0.1s with no pipeline re-run
30	[P2] Flag sarcasm-marker comments as low-confidence in the Dashboard #64	LOW	M	P2	5 sarcasm-risk comments display a low-confidence label in the Dashboard

Production Deployment Plan

Layer 1 — Frontend

Streamlit · Streamlit Cloud (free tier)

Creator uploads CSV · views sentiment, keywords, niche analysis, recommendations

No API layer

Streamlit handles frontend and backend in one process

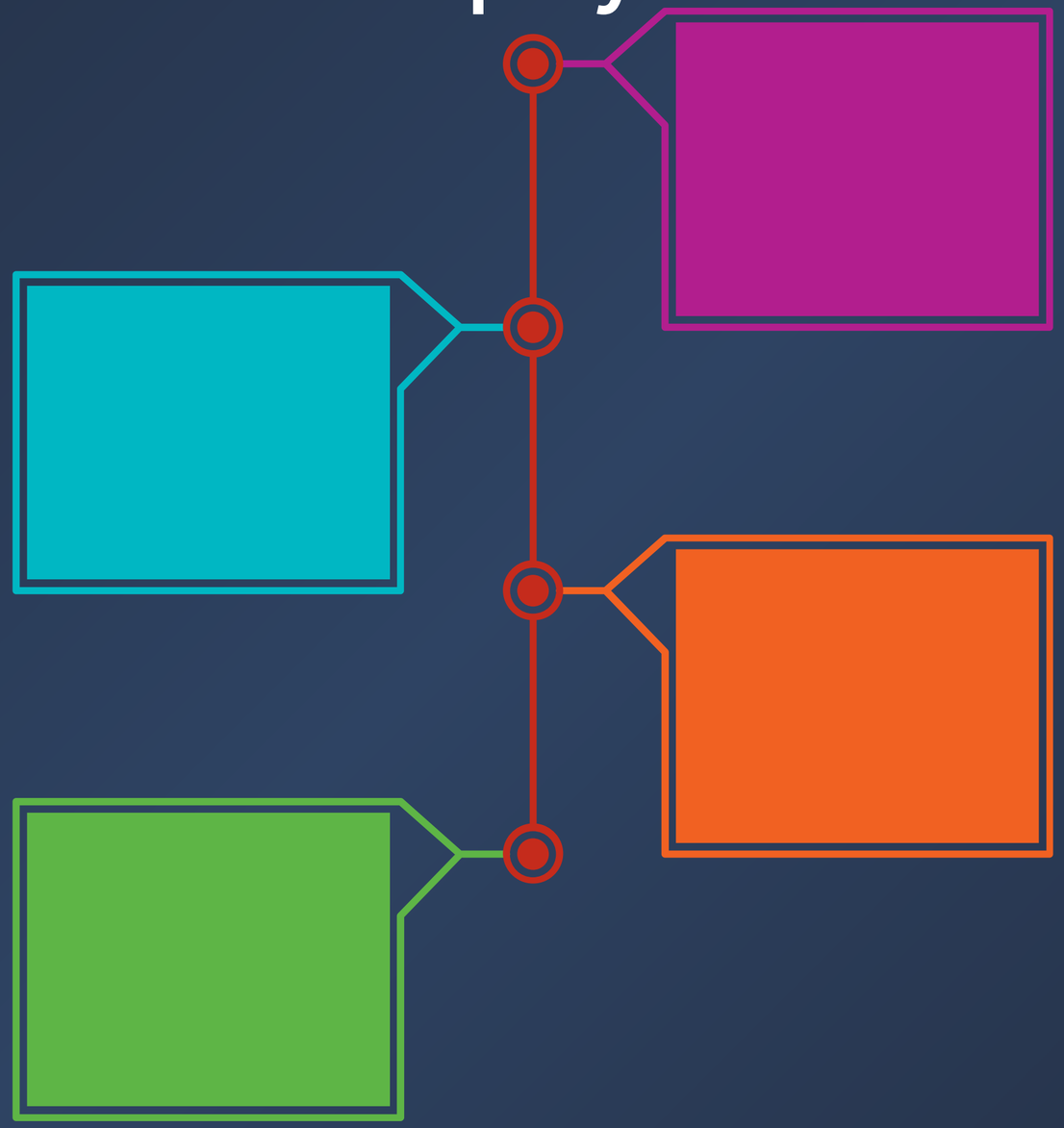
Layer 3 — NLP Pipeline (inference · fully local · no GPU needed)

Preprocessor	Cleans text · detects language (langdetect)
VADER sentiment	Scores each comment positive / neutral / negative
TF-IDF keywords	Extracts distinctive words (scikit-learn)
Niche analyzer	Compares engagement across content types
Request extractor	Finds repeated viewer asks using regex patterns

Layer 4 — Storage

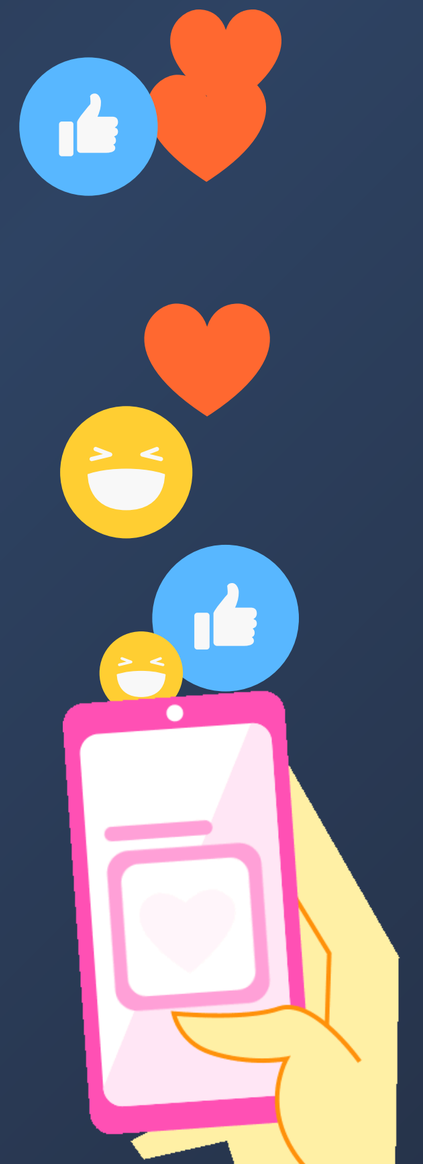
SQLite database · Streamlit session state

User accounts + analysis history · CSV held in memory, not persisted



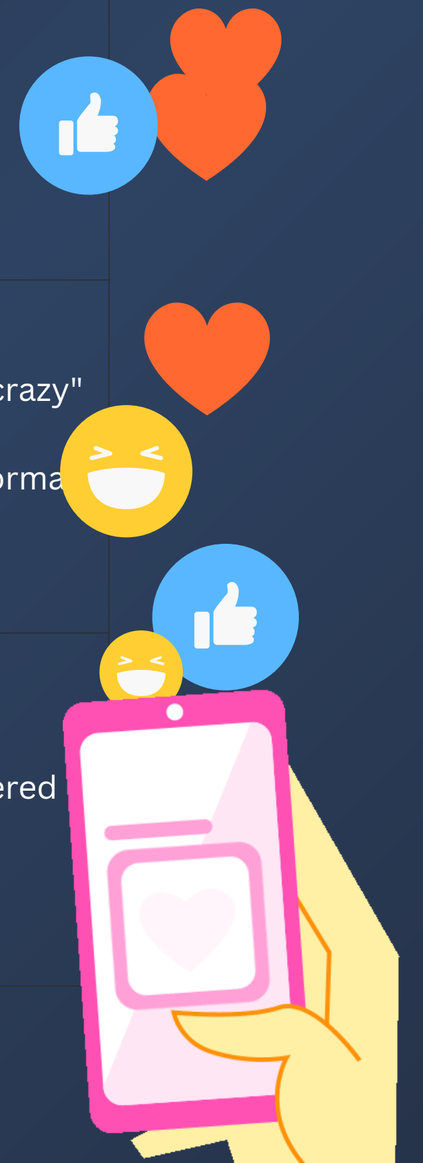
Monitoring and Observability

Metric	Target	Alert Threshold
p95 analysis latency (ms)	< 500 ms	Alert if > 1,000 ms
Non-English exclusion rate (%)	< 22%	Alert if > 30%
Upload error rate (%)	< 2%	Alert if > 5%



Task 4

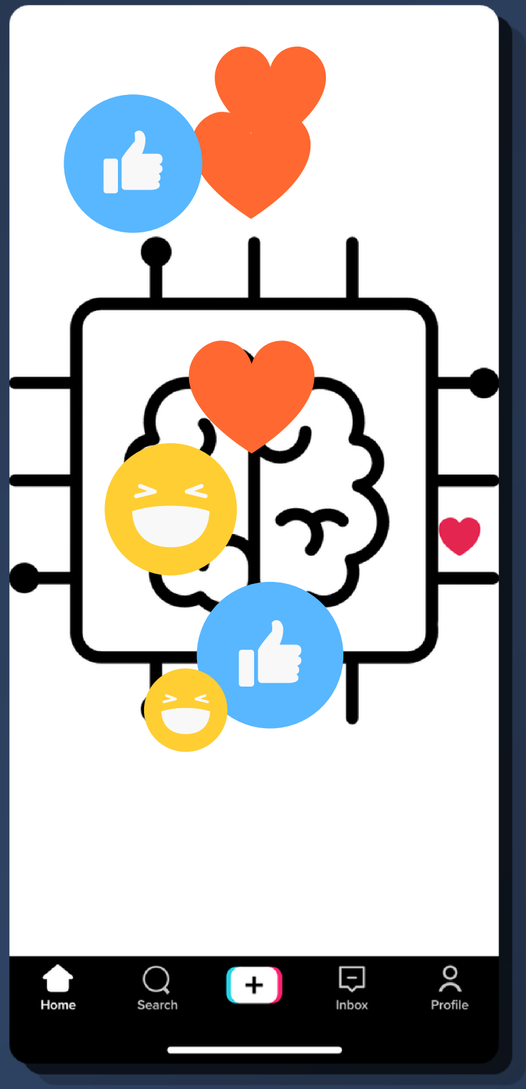
Change	Metric Before	Metric After	Verdict
Add TikTok hype slang to VADER lexicon (hard, crazy, sick, dead, fire → positive)	2/3 slang comments correctly scored	3/3 (100%)	Keep -- "This community is crazy" was scored negative, now positive; no regression on forma test set (F1 stays 94.1%)
Trust short ASCII-only comments (≤3 tokens) as English instead of dropping them	Non-English exclusion rate 22.0%	16.20%	Keep -- 71 comments recovered with no accuracy cost



task 5

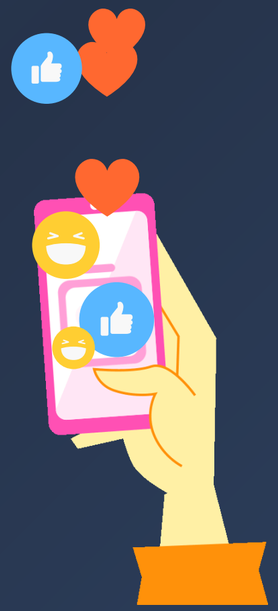
Item	Acceptance Criterion	Result
Fix langdetect false positives: trust short ASCII-only comments as English	Exclusion rate drops from 22.0% to ≤12.0%	16.2% -- PARTIAL PASS (71 comments recovered; emoji-containing short comments still excluded)
Add TikTok hype slang to VADER lexicon (hard, crazy, sick, dead, fire)	Macro F1 stays ≥94.1%; hype-slang comments score positive	F1 = 94.1% maintained; "This community is crazy" correctly rescored negative → positive -- PASS
Automate regression test via GitHub Actions CI (.github/workflows/regression.yml)	CI runs test_harness.py on every push; exits 1 if any flag fires	Workflow created; test_harness.py ran clean across 10 datasets, 0 flags -- PAS

Final Presentation



Task 1: The Hook and Narrative Arc

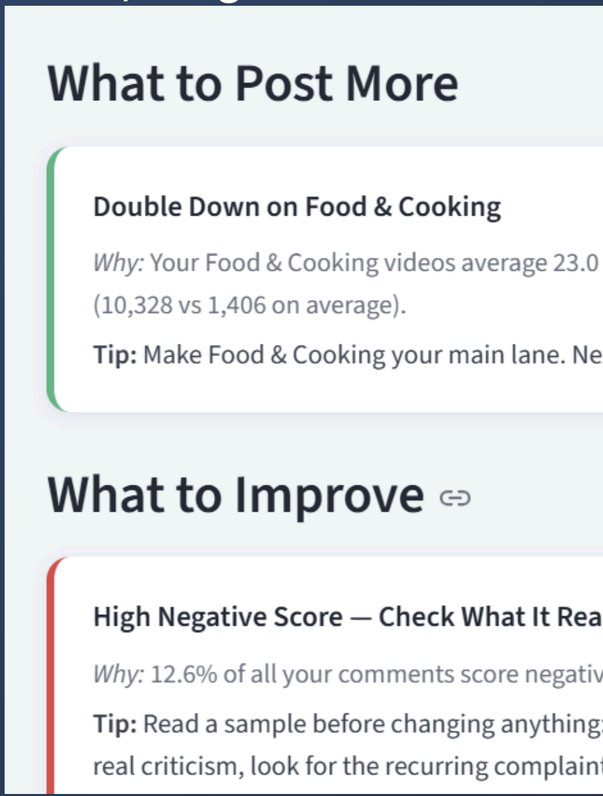
Persona
Amara, 21, student and lifestyle content creator with 2,300 TikTok followers. She posts three times a week across food, travel, and daily life content. Her pain point: she gets 50 to 100 comments per video but reads them one by one on her phone and can never tell which content type her audience actually prefers.



Before
Every Sunday evening Amara scrolls through 200+ comments across the week's videos, trying to spot patterns manually. She notices her food video got more comments than her vlog but cannot tell if that gap is real or just one lucky video. She posts whatever feels right that week and hopes for the best. Three months in, her growth has stalled and she does not know why.

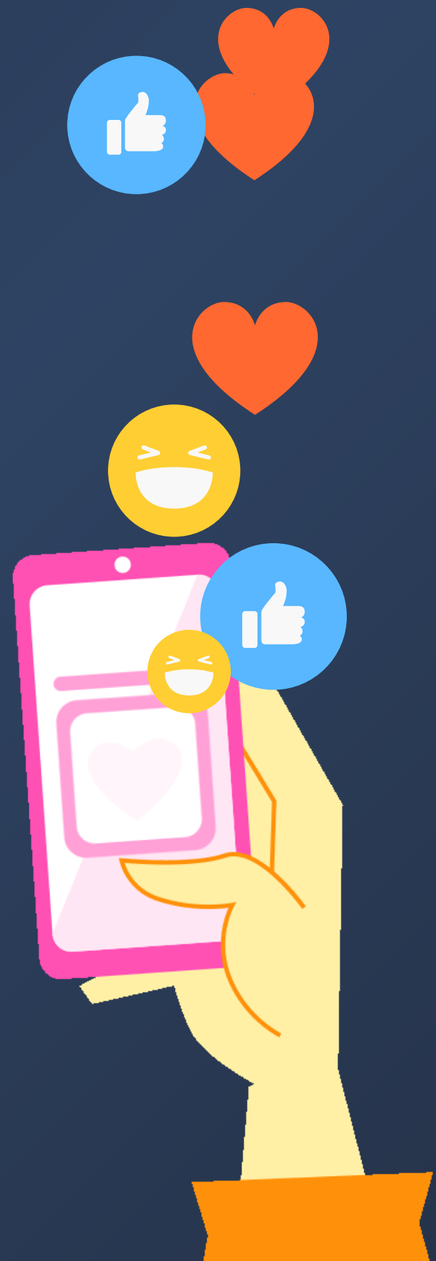
After + Elevator Pitch
Amara uploads one CSV. In under a second she sees her Food and Cooking videos average 26 comments each versus 7 for Lifestyle -- a 3.6x gap the data makes unmissable. The recommendations page shows 27 viewers already asked for a camping gear video.

Elevator Pitch: TikTok Creator Intelligence turns 200 unread comments into one clear answer about what to post next -- so small creators can grow with data, not guesswork.



Task 2: The Failure Post-Mortem

Aspect	TikTok Creator Intelligence
What failed	The recommendation system generated meaningless advice citing "content" as the top keyword -- producing outputs like "viewers are asking for content type of content"
When discovered	D8 user evaluation, Week 10 -- tester Wudh found the What to Improve section empty despite 14.4% negative sentiment, and keyword clusters showed platform words instead of topics
Root cause	TF-IDF scored generic platform words ("content", "video", "channel") highest because they appear in almost every comment -- making them statistically distinctive but meaninglessly generic; no filter existed to block them
Impact on D8 metrics	"The analysis felt true" user rating: 3.0 / 5; trust gap in keyword categorization flagged by 3 of 4 testers; negativity threshold set at 15% caused the What to Improve section to silently produce nothing for Wudh's 14.4% negative dataset
Pivot and outcome	Added GENERIC_FILLER word list to nlp/keywords.py blocking platform words from TF-IDF; rebuilt recommendation generator to only fire a card when measured evidence exists and every card cites actual numbers. Result: Wudh's issue resolved, no later tester reported generic recommendations, SUS rose to 87.5 / A+



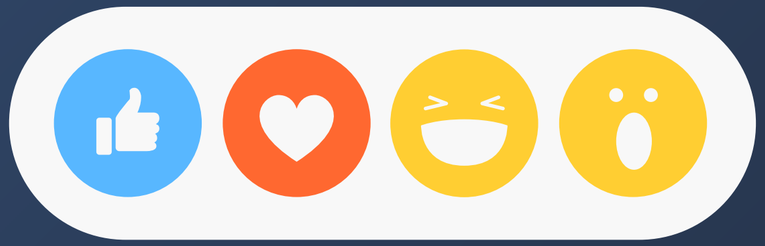
Task 3 The Live-Demo Storyboard

#	Screen State	Narration Script	Latency Risk	Backup Action
1	App open on Upload Data page, no file loaded	"Meet Amara -- 2,300 followers, posts three times a week, reads every comment on her phone and still cannot tell what is working. This is the tool I built to fix that."	None	Pre-loaded screenshot of the upload page
2	Drag and drop comments CSV into the uploader	"She exports her TikTok comments as a CSV -- one file, that is all the system needs. I am uploading my own real data from @ichbinnelo, 1,211 comments scraped across two years."	None	File already staged, single click to upload
3	Upload videos CSV into the optional uploader	"The second file is optional -- video metadata. When it is there, the system unlocks niche comparison. Without it you still get sentiment and keywords."	None	Skip this step and note the feature verbally
4	Click Run Analysis -- spinner visible for under 1 second	"One click. The pipeline runs entirely locally -- no API call, no GPU, no cost. VADER scores every comment, TF-IDF extracts the keywords, the niche analyzer compares engagement across content types."	< 0.5 s	"The analysis already finished -- watch how fast that was."
5	Dashboard page loads -- sentiment cards and charts visible	"The dashboard shows the full picture instantly. 72% positive, 14% negative. The bar chart shows which words are most distinctive in her comments -- not generic words like 'content', actual topics."	None	Pre-run screenshot of Dashboard saved on desktop
6	Navigate to Recommendations page -- Double Down card visible	"This is the part that matters. Her Food and Cooking videos average 26 comments each versus 7 for Lifestyle -- a 3.6x gap. The system spotted that without being told to look for it."	None	Read the card text aloud from memory
7	Scroll to Suggested Content Ideas section	"27 viewers asked for a camping gear video. 22 asked where a specific trail was. These are not guesses -- they are exact phrases counted across the comment section."	None	Pre-prepared list of top requests on a backup slide
8	Switch to browser tab showing GitHub Actions CI	"Every time I push code, a regression test runs automatically against 37 manually labelled comments. If accuracy drops, the push is blocked. The system protects itself."	None	Screenshot of passing CI run
9	Return to Dashboard, point to sentiment numbers	"D8 measured 91.9% accuracy and 94.1% Macro F1 against manually reviewed ground truth -- beating the naive baseline by 37 points. Small creator, real data, zero cloud cost."	None	README S8 table open in second tab

Task 4 The Lessons Learned Matrix

	Expected	Surprising
Technical	VADER works well on emoji-rich social media text -- it was designed for this, and the 91.9% accuracy on real TikTok comments confirmed the architecture choice made in D4. TF-IDF extracts keywords reliably but requires domain-specific stop words -- generic platform words ("content", "video") score highest and must be manually blocked.	Stripping emojis to "clean" the text hurt accuracy by 5.4 points -- the emoji is often the entire sentiment signal on TikTok. A 12-word GENERIC_FILLER list fixed what felt like a fundamental model problem -- the generic recommendations bug that drove the full 9-task refactor was a data filtering issue, not a model issue.
Procedural	User testing surfaces problems developers cannot see -- testers immediately found the keyword cluster opacity and generic recommendations issues that code review missed. Building an end-to-end system takes longer than isolated modules -- connecting preprocessing to sentiment to recommendations introduced bugs that never appeared in unit testing.	One tester independently found the exact VADER hype slang failure that the D8 error analysis had predicted theoretically -- user testing confirmed a documented bug without being told to look for it. The hardest deliverable was evaluation (D8), not building (D7) -- measuring quality rigorously requires more discipline than writing the code.

Most Valuable Lesson: "A pre-trained model is only as good as the data you feed it -- VADER was accurate out of the box, but every real improvement came from fixing what went in (emoji preservation, slang lexicon, generic word filtering), not from changing the model itself.



you can visiit the app here

https://tiktok-creator-intelligence.streamlit.app